

Chapter 6 – Non-Type Specific Flight Training Device – Qualification Performance Standards

6.1 Overview

- (1) This chapter describes the qualification performance standards (QPS) for aeroplane and rotorcraft Flight Training Devices (FTD) (Level 1 and 2). Wherever the term “aircraft” is used, it refers to both aeroplanes and rotorcraft. The performance standards were derived from ICAO Document 9625 Edition 3 Volume 1 and 2, with some modifications based on an analysis of single and multi-engine; multi-crew coordination and Instrument Procedures Check requirements from and CAR Part IV training program requirements. The technical requirements and evaluation criteria specific to aeroplane and rotorcraft FSTDs are presented in the following appendices and tables:

A. AEROPLANE - NON-TYPE SPECIFIC FTD - QPS

- (a) Appendix E1 – Aeroplane FTD – General Simulator Requirements
- (b) Appendix E2 – Aeroplane FTD – Objective Test Requirements
- (c) Appendix E3 – Aeroplane FTD – Simulator Subjective Evaluations

B. ROTORCRAFT - NON-TYPE SPECIFIC FTD - QPS

- (a) Appendix F1 – Rotorcraft FTD – General Simulator Requirements
- (b) Appendix F2 – Rotorcraft FTD – Objective Test Requirements
- (c) Appendix F3 – Rotorcraft FTD – Simulator Subjective Evaluations

ATTACHMENTS:

- (a) Attachment 6.1 - Sample FTD Certificate for Regional Authorities
- (b) Attachment 6.2 - Non-Type-Specific FTD Qualification Profile

6.2 Roles and Responsibilities

- (1) The NSEP is responsible for ensuring that all FSTDs used for pilot training and checking are evaluated in accordance with the requirements of CAR 606.03. To provide responsive and efficient service delivery and oversight, the MSP may delegate the conduct of initial evaluations of non-type-specific FTDs to Regional Authorities provided that:
 - (a) The FTD product or series produced by a particular manufacturer has been previously approved in by the NSEP and operated in Canada;
 - (b) The QTG does not present automated test results;
 - (c) The sponsor is not requesting additional qualifications that are not covered by the QPS in this chapter.
- (2) The MSP may delegate recurrent evaluations of Level 1 and 2 FTDs to Regional Authorities after the NSEP completes the initial evaluation and follow-up corrective action. The Regional Authority may request the NSEP to participate in the first recurrent if knowledge transfer is required (particularly in the case of automatic testing).
- (3) The Regional Authority assuming oversight of a Level 1 or 2 FTD is responsible for:
 - (a) verifying FTD sponsor eligibility;
 - (b) conducting initial evaluations as delegated;

- (c) conducting recurrent evaluations in accordance with CAR 606.03;
 - (d) tracking and ensuring corrective action of discrepancies noted during the evaluations;
 - (e) Issue certificate:
 - (f) TCID number,
 - (g) FTD Sponsor,
 - (h) FTD location,
 - (i) manufacturer/model,
 - (j) aircraft class/set simulated,
 - (k) serial number,
 - (l) issuing authority (Region)
 - (m) FTD Level and special qualifications,
 - (n) QPS used
 - (o) Date of issue and expiry date.
 - (p) Qualification Profile (as page 2).
- (4) Maintaining a record of FTD certificates, TCID numbers issued; evaluation reports and associated qualification documents.

Note: Sample certificate and qualification profile form is shown at attachments 6.1 and 6.2 (RDIMS #13698343).

Note: TCID numbers issued by Regional Authorities are 4 digits, where the first digit represents the Region as follows: Atlantic Region – 2xxx; Quebec Region - 3xxx; Ontario Region - 4xxx; Prairie Northern Region – 5xxx; Pacific Region – 6xxx. TCID numbers 0 to 1999 are reserved for NSEP use.

6.3 General Description of Non-Type-Specific FTDs

- (1) Unlike type-specific FTDs, non-type-specific FTDs are intended to be representative of a class or set of aircrafts. The expression “generic device” has traditionally been used to designate such non-type-specific devices; however, in this chapter, the expression “generic device” has been replaced by “non-type-specific” to preclude confusion with the simulation feature fidelity as defined by the Table 6.2-1 below.

Note: Class” in relation to the classification of aircrafts, means aircrafts having similar operating characteristics to single-engine aircrafts, multi-engine aircrafts, centre-line thrust aircrafts, land aircrafts or sea aircrafts.

Note: Set of Aircraft” is a group or series of aircraft group that share similar handling and operating characteristics, similar operating envelopes, and have the same number and type of engines or power-plants.

Fidelity	Aircraft simulation	Sound, Motion and Visual Cueing	Environment simulation
Generic	Not specific to aircraft model, type or variant.	Generic to an aircraft of its class. Simple modelling of key basic cueing features. For visual cueing only: Generic visual environment with perspective	Simple modelling of key basic environment features.

Fidelity	Aircraft simulation	Sound, Motion and Visual Cueing	Environment simulation
		sufficient to support basic instrument flying and transition to visual from straight in instrument approaches.	
Representative	Representative of an aircraft of its class, e.g., to engine turbo-prop aircraft. It does not have to be type specific.	For sound and motion cueing only: Replicates a specific aircraft in set of aircrafts to the maximum extent possible. For visual cueing only: Representative of the real world visual environment and perspective.	Representative of the real world environment.
Specific	Replicates the specific aeroplane model, type or variant.	Applicable to visual cueing only: replicates the real world visual environment and (infinity) perspective.	Replicates the real world environment as far as possible for any specific location.

6.4 Type-Specific FTD Levels

- (1) The Non-Type-Specific FTD levels were developed to provide guidance and minimum FTD requirements to Flight Training Units (FTUs) who seek approval to use a lower cost FTDs in approved training programs for single and multi-engine licensing, multi crew coordination (MCC) (for aeroplanes only), and Instrument Proficiency Checks (IPC) (for aeroplanes only). These devices are not authorized for any experience, training or checking credits in type-specific approved training programs. MCC and IPC training or checking on FTDs simulating aeroplanes that require a type-rating may only be conducted on type-specific FTDs (See para 6.4.3 for eligible FTDs)
- (2) There are two basic non-type-specific FTD configuration:
 - (a) **Level 1** - A level 1 FTD is the lowest fidelity device and may have a “generic” cockpit design. The device does not require a visual system or active force-feedback flight controls. It will support basic instrument procedural training, and may provide experience credit for 5 hours pursuant to CAR 421.26(4)(a) pending evaluation.
 - (b) **Level 2** – A Level 2 FTD, is a device that represents a “designated” class and set of aircraft. The device requires active force-feedback flight controls and requires a visual system that meets the minimum requirements in Tables E1A or F1A as applicable. It will support basic instrument procedure training and may provide experience credit for 10 hours pursuant to CAR 421.30(4)(a)(ii)(A)(III) pending evaluation.
- (3) An aeroplane FTD may qualify for instrument proficiency checks (IPC) and/or for multi-crew coordination (MCC) training in an ATP(A) program if the device meets the following criteria.
 - (a) MCC Qualification. An FTD may qualify as a Multi-Crew Coordination FTD provided that it:
 - (i) Meets the criteria objective and subjective test criteria for an FTD Level 2, 5, 6, or 7; and

- (ii) Meets or exceeds the criteria outlined in Appendices E for the MCC qualification; and
 - (iii) Fully supports the procedural and operational performance tasks required in the TC approved training program for ATP(A), per CAR as amended.
- (4) IPC Qualification. An FTD will qualify as a platform to conduct Instrument Proficiency Checks (IPC) provided that it:
 - (a) Meets the criteria objective and subjective test criteria for an FTD Level 2, 5, 6, or 7;
 - (b) Meets or exceeds the criteria outlined in Appendices E for the IPC qualification; and
 - (c) Fully supports the procedural and operational performance tasks IPC in the Advisory Circular AC-401-004 as amended to conduct an IPC.

Note: An aeroplane FTD granted an additional MCC or IPC qualification will be designated on the FSTD certificate as Level #-MCC. (i.e. Level 5-MCC; Level 2-IPC; Level 6-MCC/IPC)

Note: The term “Designated” is used when a feature or requirement must be specifically identified by the sponsor prior to the initial evaluation or update evaluation.

- (5) Detailed standards and tests for the various levels of FTDs are fully defined in Appendices E for aeroplanes and Appendices F for rotorcraft.

6.5 General Design Standards

- (1) To avoid any delays in the design, build, acceptance, or qualification process, the sponsor should engage in early discussions with the TPAA and Approving Authority to broadly agree on the “designated aircraft configuration” and any additional qualifications (MCC or IPC) intended.
- (2) Cockpit/Flight Deck Configuration. The Sponsor should choose a configuration that sensibly represents the aircrafts likely to be used in the approved training program. To ensure a good training environment, the cockpit/flight deck should be located in a training environment that minimizes any distractions.
 - (a) **Level 1.** The cockpit/flight deck must sufficiently represent a class of aircraft to support approved use. The cockpit/flight deck may be open or enclosed. The appearance, arrangement, operation, and function of physical switches and other required controls must be realistically placed. The instrument panel of an aeroplane FTD must include at least the following:
 - (i) Master/battery;
 - (ii) Magnetos for each engine (as applicable);
 - (iii) Alternators for each engine;
 - (iv) Fuel boost pumps for each engine;
 - (v) Avionics master;
 - (vi) Pitot heat; and
 - (vii) Rotating beacon/strobe, navigation, taxi, and landing lights.
 - (b) **Level 2.** The cockpit/flight deck must be representative of a set of aircrafts or the aircraft itself to support approved use. The cockpit/flight deck must be sufficiently enclosed to minimize distractions. The physical arrangement, appearance, and operation of controls, instruments, and switches should model at least one aircraft in the family of aircraft represented as closely as practicable.

- (3) **Cockpit/Flight Deck Components.** Physical flight and aircraft system controls must be recognizable as to their function and how they are to be manipulated solely from their appearance.
- (a) Physical flight and aircraft system controls must not use interfaces such as a keyboard, mouse, or gaming joystick to control the represented aircraft in simulated flight.
 - (b) Components used within the cockpit/flight deck area do not need to be aircraft parts; however, any parts used should be representative of typical training aircrafts and should be robust enough to endure the training tasks.
 - (c) The use of simple flat display technology-based representations and touch-screen controls to represent objects other than basic push-button types of controls will not be acceptable. The training tasks envisaged for these devices are such that appropriate layout and feel is very important. For example, the altimeter sub-scale knob needs to be physically located on the altimeter. The use of flat display technologies with physical overlays incorporating operational switches/knobs/ buttons replicating an aircraft instrument panel may be acceptable.

6.6 Data package

- (1) The data for the aerodynamics model, flight controls and engines used in Level 1 or 2 FTDs must be based on sound engineering principles and substantiated in the form of a “Reference Data” engineering report (RDR). The RDR shall show that the proposed validation data are representative of the aircraft or the class of aircrafts modelled. This report may include flight test data, manufacturer’s design data, information from the aircraft flight manual (AFM) and maintenance manuals, results of approved or commonly accepted simulations or predictive models, recognized theoretical results, information from the public domain, or other sources as deemed necessary by the FTD manufacturer to substantiate the proposed model.
- (2) Generic or representative data may be derived from a specific aircraft within the class of aircrafts simulated, or it may be based on information from several aircrafts within a group of aircrafts that reflect the performance of the “designated aircraft configuration”.
- (3) With the concurrence of the NSEP, the data may be in the form of a manufacturer’s previously approved set of validation data for the applicable FTD. The sponsor should submit the intended validation data together with a substantiation report to the NSEP for review and approval prior to procurement or the commencement of the manufacturing or procurement process. Once the set of data for a specific FTD has been accepted and approved by the NSEP, it will become the validation data that will be used as reference for subsequent recurrent evaluations with the application of the stated tolerances.
- (4) **Data Collection and Model Development.** A basic requirement for any modelling is the integrity of the mathematical equations and models used to represent the flying qualities and performance of the aircraft or class of aircrafts being simulated. The models should be continuous and vary sensibly throughout the required training flight envelope. The following principles apply:
 - (a) Additional data to refine the non-type-specific model can be obtained from many sources, such as aircraft design data, flight and maintenance manuals, observations on the ground and in the air, etc., without necessarily having to conduct expensive, dedicated flight testing.
 - (b) Data obtained on the ground and in flight can be measured and recorded using a range of simple means such as video cameras, paper and pencil, stopwatch, and new technologies (e.g. GPS).
 - (c) Data gathering should take place at representative masses and centres of gravity. Development of such a data package including justification and the rationale for the design and intended performance, the measurement methods and recorded parameters

(e.g. mass, CG, atmospheric conditions) should be carefully documented and available for inspection by the NSEP as part of the qualification process.

- (5) Flight Controls. There can be a strong interaction between the flight control forces and the effects of both the engines and the aerodynamic configuration and will reflect the fidelity and credits offered to the FTD:
 - (a) **Level 1** FTDs only require “generic” fidelity level flight controls, which may be achieved with a passive force cueing system utilizing springs. The evaluator will assess the fidelity of the flight controls during the initial and recurrent evaluations. Qualification will not be awarded if the flight controls manifest any characteristics that would result in negative training.
 - (b) **Level 2** FTDs require “representative” fidelity level flight controls, which will require an active force feedback cueing system that allows flight controls forces to vary with position as well as aircraft configuration (speed, flaps, trim).

6.7 Visual System

- (1) The adequacy of the performance of the visual system will be determined by its ability to support the intended training tasks, e.g. “visual cueing sufficient to support changes in approach path by using runway perspective”.
- (2) Collimated visual optics are not required for non-type-specific FTDs since no training credits for landing are available. A single channel direct viewing system is acceptable for level 2. For MCC and IPC additional qualifications, the visual FOV must be as per Appendix E2 (applicable to aeroplanes FTDs only).
- (3) Where the device does not simulate a particular aircraft type, the design of the out-of-cockpit/flight deck view should be matched to the visual system so that the pilot has a FOV sufficient for the intended training tasks. For example, during an instrument approach the pilot should be able to see the appropriate visual segment at decision height. Additionally, where the aircraft deviates from the normal approach path, undue loss of visual reference should not occur during the subsequent correction.

6.8 Transport Delay

- (1) For a non-type-specific device, a transport delay test may be used to demonstrate that the FSTD system does not exceed the permissible delay. However, for such a device using simple models, a Statement of Compliance (SOC) may be acceptable in lieu of a test. The SOC must contain a system block diagram that support a description of the cumulative latency or transport delay from control input to instrument, visual and motion system response as applicable. The SOC should also describe the method of determining the transport delay.
- (2) The maximum permissible transport delays is 300 milliseconds for aeroplanes and 120 milliseconds for rotorcraft. A test may be required during the initial evaluation to verify the SOC if the calculated transport delay is close to the maximum permissible value.

6.9 Testing/Evaluation

- (1) To ensure that any device meets its design criteria initially and periodically throughout its life, a system of objective and subjective testing will be used.
- (2) The validation tests specified in Appendix E2 for aeroplanes and Appendix F2 for rotorcrafts will be flown by a suitably skilled person and the results recorded manually.
- (3) **Automatic Recording and Testing.** The MSP must be notified if a sponsor wishes to use automatic recording and testing for validation tests. In such a case, the QTG review will be

conducted by the NSEP. Based on the complexity of the QTG and the results of the QTG review, the MSP will decide whether it is appropriate to retain NSEP oversight of the device or to delegate oversight to the Regional Authority.

- (4) The tolerances specified are designed to ensure that the device meets its original target criteria year after year. It is therefore important that such target data are carefully derived and values are accepted by the NSEP or Regional Authority in advance of the initial evaluation (QTG review). Unlike the target parameters stipulated for type-specific devices, those stated for non-type-specific FSTDs are primarily intended to ensure repeatability during the life of the FSTD and in particular at each recurrent regulatory inspection. For initial qualification, the device should meet its design criteria within the listed tolerances; however, the evaluator may approve reasonably altered target values if they do not adversely affect the performance of the FTD or the quality of training and the results are repeatable.
- (5) A number of tests within the QTG have had their tolerances reduced to “Correct Trend and Magnitude” (CT&M) thereby avoiding the need for specific validation data. The use of CT&M is not to be taken as an indication that certain areas of simulation can be ignored. For such tests, the performance of the FSTD should be appropriate and representative of the simulated aircraft and should under no circumstances exhibit negative characteristics. Where CT&M is used as a tolerance, it is strongly recommended that a simple, reliable, and repeatable test method be designed to establish a baseline result approved during the initial evaluation. This will provide a means of avoid or resolving any divergent subjective opinions during recurrent evaluations.
- (6) **Longitudinal change force tests.** For Level 1 and 2 aeroplane FTDs, it is acceptable to use change force tests instead of the longitudinal power, flap and gear change dynamics tests. The purpose of these change force tests is to simply reproduce the control force feeling during recurrent evaluations. These tests should be conducted in the following way:
 - (a) Trim the aeroplane in straight and level flight in the most suitable configuration;
 - (b) Initiate the configuration change (power, flap or gear change);
 - (c) Maintain the airspeed constant using the pitch control (or as an alternate procedure, maintain the altitude of the original trimmed conditions); and
 - (d) Measure the required pitch control force change.

Note: If automatic testing is used, a change dynamics test is acceptable.

- (7) The subjective tests listed in Appendix E (aeroplanes) and F (rotorcraft) must be flown by a suitably qualified and experienced pilot.

6.10 Initial and Upgrade) Qualification Requirements

- (1) In order to be qualified at a particular qualification level, the FTD must:
 - (a) Meet the general requirements at Appendix E1 for aeroplanes, or Appendix F1 for rotorcrafts;
 - (b) Meet the objective testing requirements at Appendix E2 for aeroplanes, or Appendix F2 for rotorcrafts; and
 - (c) Satisfactorily accomplish the subjective tests at Appendix E3 for aeroplanes, or Appendix F3 for rotorcrafts.
- (2) Sponsors applying for an initial evaluation must follow the procedures at Section 6.11 of this chapter. The evaluating agency must be in receipt of the following prior to conducting an evaluation:
 - (a) A statement that the FTD meets all provisions of the QPS. The Sponsor's application must also:

- (i) identify the class, set of aircraft or aircraft type simulated and any additional qualifications requested (i.e. IPC or MCC): and
 - (ii) Designate airport navigation and visual scenes, and operational systems as required by the QPS for the level and additional qualifications requested.
- (b) Unless otherwise authorized, the sponsor must send a letter of compliance to the MSP or Regional Authority (as described in paragraph 6.11.1c of this chapter) no later than 5 business days prior to the scheduled.
- (c) A QTG, acceptable to the MSP, that includes all of the following where required by the QPS:
 - (i) Objective data obtained from aircraft testing or another approved source.
 - (ii) Objective test results obtained from the performance of the FTD.
 - (iii) Result of FTD subjective tests.
 - (iv) A description of the equipment necessary to perform the evaluation for initial qualification and the continuing qualification evaluations.
- (d) Unless otherwise approved by the MSP, the sponsor's QTG submission must include results of all required tests conducted at the sponsor's training facility after the FTD is assembled with systems and sub-systems functional and operating in an interactive manner.
- (3) During the initial (or upgrade) qualification evaluation, the sponsor must provide a person to assist with the evaluation who is a user of the device (e.g., a qualified pilot or instructor pilot with flight time experience in that aircraft) and knowledgeable about the operation of the aircraft and the FTD.

6.11 Application for initial or Upgrade Evaluations

- (1) The application process for an initial or upgrade qualification evaluation for a non-type-specific FTD will be administered according to the following steps:
 - (a) **Letter of Intent.** For each FSTD, the sponsor must submit a "letter of intent" requesting an initial qualification evaluation to their Regional Authority no later than 120 days prior to the evaluation date requested. The letter will include:
 - (i) A description of the FSTD;
 - (ii) The anticipated qualification level of the FSTD;
 - (iii) The make, model, and series of aircraft simulated; and any other pertinent information; and
 - (iv) Any additional information that may affect the scope of the evaluation (i.e. FSTD relocation, sponsorship change, restoration after voluntary suspension etc.)

Note: To avoid any qualification delays, the sponsor should contact the MSP as early as possible to coordinate a review and acceptance of the validation data package.

- (2) **Regional Authority's Response.** The Regional Authority will liaise with the MSP to determine the most appropriate evaluation agency (NSEP or Regional Authority). The designated evaluating agency will reply to the letter of intent within 15 business days with the following information:
 - (a) Tentative dates for the initial evaluation;
 - (b) The date the sponsor must submit the QTG if applicable;

- (c) An attached “qualification profile” form to be completed and returned with the QTG submission.
 - (d) Any special equipment and qualified personnel needed to accomplish or assist in the accomplishment of tests during initial qualification.
- (3) **QTG Review.** The Sponsor will submit the QTG and the Reference Data Report (RDR) in a form and manner described in this chapter. The evaluating agency will review the QTG and forward comments and questions resulting from the review to the Sponsor for their response. Depending on the results of the review, the designated evaluating agency will confirm or postpone the evaluation date. The sponsor will provide a response to the QTG review comments no later than 5 working days before the evaluation date.
- (4) **Letter of Compliance.** The Sponsor will submit a “letter of compliance” signed by the management representative no later than 5 working days before the evaluation confirming that:
- (a) Qualified pilot(s), after having flown all of the applicable subjective evaluation in the applicable QPS appendix tests (Table E1B for aeroplane FTDs) relevant to the qualification level of the FSTD, have determined that the performance and handling qualities of the FTD represent those of the class or set of aircraft within the normal operating envelope. Exceptions, if any, must be noted. The name of the pilot(s) making this determination must be identified on the letter of compliance. Exceptions, if any, must be noted.
 - (b) Qualified pilot(s) or person(s) trained on simulator systems/sub-systems and trained on the operation of the simulated aircraft systems, after having exercised the operation of the FTD and the pertinent functions available through the Instructor Operating Station(s), have determined that the FTD cockpit configuration, systems, and sub-systems (including the simulated aircraft systems) functionally represent those in the aircraft class or set of aircraft. The name of the person(s) making this determination must be available to the MSP upon request. Exceptions, if any, must be noted.
 - (c) When the FSTD passes an evaluation for initial qualification, the MSP issues a Qualification Certificate as described at paragraph 6.2.3 of this chapter.
- (5) After the NSEP completes the evaluation for initial qualification, the sponsor must update the Qualification Test Guide (QTG), with the results of the TC-witnessed tests as determined by the evaluator.
- (6) Upon issuance of the Qualification Certificate the updated QTG becomes the Master Qualification Test Guide (MQTG). The MQTG must be made available to the Regional Authority or NSEP upon request.

6.12 Qualification Test Guide (QTG)

- (1) The sponsor must submit the QTG to the NSEP or delegated Regional Authority for review and approval, and must include:
- (a) A QTG cover page with sponsor and TC approval signature blocks;
 - (b) An FTD information page that provides the information listed in this. For convertible FTDs, the sponsor must submit a separate page for each configuration of the FTD;
 - (i) The sponsor's FTD identification number or code.
 - (ii) The aircraft model and series being simulated.
 - (iii) The reference to the data record document.
 - (iv) The source of the basic aerodynamic model and the aerodynamic coefficient data used to modify the basic model.

- (v) The engine model(s) (class)
 - (vi) The flight control configuration (reference aircraft model).
 - (vii) The flight management system identification and revision level.
 - (viii) The FTD model and manufacturer.
 - (ix) The date of FTD manufacture.
 - (x) The FTD computer identification.
 - (xi) The visual system model and manufacturer, including display type.
 - (xii) The motion system type and manufacturer, including degrees of freedom.
- (2) A Table of Contents.
 - (a) A log of revisions and a list of effective pages.
 - (b) List of all relevant data references.
 - (c) A glossary of terms and symbols used (including sign conventions and units).
 - (d) Statements of compliance and capability (SOCs) as required by the QPS.
 - (e) Recording procedures or equipment required to accomplish the objective tests if applicable.
 - (f) The following information for each objective test required in Appendix E2, Table E2A (aeroplanes) or Appendix F2, Table F2A (rotorcraft) as applicable:
 - (g) Name of the test.
 - (h) Objective of the test.
 - (i) Tolerances parameters and flight conditions.
 - (j) Pertinent and complete Manual test procedures. (Including a list of all relevant parameters driven or constrained during the manually test setup).
 - (k) A means of comparing the FTD test results to the reference data or targets.
 - (l) Other information as necessary, to assist in the evaluation or interpretation of the test results.
- (3) For FTDs using **Auto-testing utilities**, the following information in addition to para i (above) must be included for each objective test:
 - (a) Initial conditions.
 - (b) Automatic test procedures (if applicable).
 - (c) Method for evaluating FTD objective test results.
 - (d) A list of all relevant parameters driven or constrained during the automatically conducted test(s).
 - (e) Tolerances for relevant parameters.
 - (f) Source of Validation Data (document and page number).
 - (g) Copy of the Validation Data (if located in a separate binder, a cross reference for the identification and page number for pertinent data location must be provided).
 - (h) Simulator Objective Test Results as obtained by the sponsor. Each test result must reflect the date completed and must be clearly labeled as a product of the device being tested.

- (4) TC considers convertible FTDs as a separate FTD for each converted configuration. If a sponsor seeks qualification for two or more models of an aircraft type using a convertible FTD, the sponsor must submit a QTG for each aircraft model, or a QTG for the first aircraft model and a supplement to that QTG for each additional aircraft model. The evaluating agency will conduct a separate evaluation on each aircraft model and issue the sponsor a separate TCID / certificate for each model that satisfactorily meets evaluation criteria.
- (5) The following describes the acceptable form and manner of presentation of objective test results in the QTG:
 - (a) The sponsor's FTD test results must be recorded and presented in a manner that allows easy comparison of the FTD test results to the validation data or targets.
 - (b) Tests results must be labeled using terminology common to aircraft parameters as opposed to computer software identifications.
 - (c) Validation data documents included in a QTG may be photographically reduced only if such reduction will not alter the graphic scaling or cause difficulties in scale interpretation or resolution.
 - (d) Scaling on graphical presentations must provide the resolution necessary to evaluate the parameters shown in Appendix E2, Table E2A for aeroplanes, or Appendix F2, Table F2A for rotorcrafts.
 - (e) For auto-tests that involve time histories, data sheets (or transparencies thereof) and FTD test results must be clearly marked with appropriate reference points to ensure an accurate comparison between the FTD and the aircraft with respect to time. Time histories recorded via a line printer are to be clearly identified for cross plotting on the aircraft data. Over-plots must not obscure the reference data.
- (6) The sponsor must maintain a copy of the MQTG at the FTD location.

Note: A sample QTG for Level 1 and 2 FTDs can be found in TP13799 - Qualification Test Guide – CAR Part IV Flight Training Devices.

6.13 Notification of FTD Modifications and updates

- (1) The sponsor must notify the Regional Authority in writing at least 21 days prior to making a modification to a non-type-specific FTD. For the purposes of this manual, an FSTD is said to have been modified when:
 - (a) Equipment or devices intended to simulate aircraft appliances are added to or removed from the FSTD, which change the MQTG or the Qualification Certificate issued by the Minister; or
 - (b) Changes are made to either software or hardware that;
 - (i) are intended to impact flight or ground dynamics;
 - (ii) that impact performance or handling characteristics of the FSTD (including motion, visual, control loading, or sound systems for those FSTD levels requiring sound tests and measurements); or
 - (iii) require an amendment to the MQTG.

Note: Routine software or hardware changes, or changes to the MQTG which do not affect required objective testing results or validation data approved during the initial evaluation of the FSTD are not considered modifications.

- (2) **Notification and approval process.** The notification and approval process for an update or modification is as follows:

- (a) **Letter of Intent.** For each FTD, the sponsor will submit a “letter of intent” to the Regional Authority that:
 - i) Provides a description of the modification or update;
 - ii) Identifies any procedural, functional or physical change to the FSTD that may affect pilot training and checking;
 - iii) Identifies impact flight or ground dynamics; changes are made that impact performance or handling characteristics of the FTD;
 - iv) Identifies any required changes to the MQTG; and
 - v) Provides a date when the modification or update would be available for evaluation.
- (b) **Regional Authority Response.** The regional office will reply to the letter of intent within 5 business days with the following;
 - i) Whether an update evaluation is required,
 - ii) Tentative date and scope of the update evaluation if required,
 - iii) Any special equipment and qualified personnel needed to accomplish or assist in the accomplishment of tests during the update evaluation, and
 - iv) Any additional requirements for approval if an update evaluation is deemed unnecessary.

Note: Depending on the complexity and impact on training, the MSP may approve a modification based on a letter of compliance from the sponsor.

- (c) **Letter of Compliance.** The Sponsor will submit a “letter of compliance” signed by the management representative no later than 5 working days before a modification is implemented or evaluated confirming that:
 - (i) Qualified pilot(s), after having flown the FTD, have determined that the performance and handling qualities represent those of the class or set of aircraft within the normal operating envelope. Exceptions, if any, must be noted. The name of the pilot(s) making this determination must be identified on the letter of compliance. Exceptions, if any, must be noted.
 - (ii) Qualified pilot(s) or person(s) trained on simulator systems/sub-systems and trained on the operation of the simulated aircraft systems, after having exercised the operation of the FTD and the affected functions through the Instructor Operating Station(s), have determined that all affected systems, and sub-systems (including the simulated aircraft systems) functionally represent those in the aircraft class or set of aircraft. The name of the person(s) making this determination must be available to the MSP upon request. Exceptions, if any, must be noted.
- (3) **Approval.** The Regional Authority will approve the use of a modification when:
- (a) The FTD successfully passes a special evaluation; or
 - (b) The Regional Authority has notified the sponsor that the letter of compliance has been accepted in lieu of an evaluation.; and
 - (c) The Qualification Certificate has been appropriately amended to reflect any changes to the FTD.

6.14 Conduct of Initial and Update Evaluation

- (1) The Regional Authority may only evaluate non-type-specific FTDs for sponsors holding a TC certificate as a Flight Training Unit. The NSEP will evaluate all type-specific FTDs.
- (2) The evaluator will assess each FTD configuration as completely as possible. To ensure a thorough and uniform evaluation, each FTD will be evaluated against the general simulator requirements in Appendix E1 or F1, the objective tests listed in Appendix E2 (aeroplanes) or F2 (rotorcraft), and the subjective tests listed in Appendix E3 (aeroplanes) or F3 (rotorcraft), as applicable. The evaluations will include, but not necessarily be limited to the following:
 - (a) Aircraft responses, including longitudinal and lateral-directional control responses (see Appendix E2 (aeroplanes) or F2 (rotorcraft) as applicable);
 - (b) Performance in authorized portions of the simulated aircraft's operating envelope, to include tasks evaluated by the MSP in the areas of surface operations, takeoff, climb, cruise, descent, approach, and landing as well as abnormal and emergency operations (see Appendix E2 (aeroplanes) or F2 (rotorcraft));
 - (c) Control checks (see Appendix E1 and E2 (aeroplanes) or Appendix F1 and F2 (rotorcraft) as applicable);
 - (d) Flight deck configuration (see Appendix E1 (aeroplanes) or F1 (rotorcraft) as applicable);
 - (e) Pilot, flight engineer, and instructor station functions checks (see Appendix E1 and E3 (aeroplanes) or Appendix F1 and F3 (rotorcraft) as applicable);
 - (f) Aircraft systems and sub-systems (as appropriate) as compared to the aircraft simulated (see Appendix E1 and E3 (aeroplanes) or Appendix F1 and F3 (rotorcraft) as applicable);
 - (g) FTD systems and sub-systems, including force cueing (motion), visual, and aural (sound) systems, as appropriate (see Appendix E1 and F2 (aeroplanes) or Appendix E1 and F2 (rotorcraft) as applicable); and
 - (h) Certain additional requirements, depending upon the qualification level sought, including equipment or circumstances that may become hazardous to the occupants. The sponsor may be subject to Occupational Safety and Health Administration requirements.
- (3) The evaluator will administer the objective and subjective tests, which includes an examination of functions. The tests include a qualitative assessment of the FTD by a TC pilot.
 - (a) Objective tests provide a basis for measuring and evaluating FTD performance and determining compliance with the requirements of this chapter.
 - (b) Subjective tests provide a basis for:
 - (i) Evaluating the capability of the FTD to perform over a typical utilization period;
 - (ii) Determining that the FTD satisfactorily simulates each required task;
 - (iii) Verifying correct operation of the FTD controls, instruments, and systems; and
 - (iv) Demonstrating compliance with the requirements of this manual.
- (4) In addition to the scheduled continuing qualification evaluation, each FTD is subject to evaluations conducted by the NSEP or Regional Authority at any time without prior notification to the sponsor. Such evaluations would be accomplished in a normal manner (i.e. requiring exclusive use of the FTD for the conduct of objective and subjective tests and an examination of functions) if the FTD is not being used for pilot training and checking. However, if the FTD were being used, the evaluation would be conducted in a non-exclusive manner. The FTD evaluator will accompany the instructor aboard the FTD along with the student(s) and observe the operation of the FTD during the training, testing, or checking activities in a non-interference manner.

- (5) If it is determined that the results of an objective test do not support the level requested but do support a lower level, the Regional Authority may qualify the FTD at that lower level or remove an additional qualification to permit the device to enter into training.
- (6) After an FTD is evaluated and deemed compliant, the Regional Authority issues a Qualification Certificate to the sponsor. The Certificate is also the Regional Authority's recommendation to the TPAA that the FTD may be used in a TC approved training program.
- (7) The numbering system used for objective test results in the QTG should closely follow the numbering system set out in Appendix E2, Table E2A (aeroplanes) or Appendix F2, Table F2A (rotorcraft) as applicable.

6.15 Continuing Qualification Evaluations and Requirements

- (1) The sponsor must conduct objective tests at least once annually and retain test results and the FTD defect log for at least two consecutive years.
- (2) During the continuing qualification evaluation conducted by the Regional Authority, the sponsor must also provide a person knowledgeable about the operation of the aircraft and the operation of the FTD.
- (3) The Regional Authority will ensure the frequency of continuing qualification evaluations is in accordance with the requirements of CAR 606.03 unless the Regional Authority becomes aware of discrepancies or performance problems with the device that warrants more frequent evaluations.
- (4) The continuing qualification evaluations will normally require 4 hours of FTD time. The sponsor should anticipate that evaluations will consist of the following:
 - (a) Review of the sponsor-conducted objective tests results and defect log.
 - (b) A random selection of objective tests from the MQTG that provide an adequate opportunity to evaluate the performance of the FTD. The tests chosen will be performed manually and should be able to be conducted within approximately one-third of the allotted FTD time.
 - (c) A subjective evaluation of the FTD to perform a representative sampling of the tasks set out in Appendix E3 (aeroplanes) or Appendix F3 (rotorcraft) as applicable. This portion of the evaluation should take approximately two-thirds of the allotted FTD time.
 - (d) An examination of the functions of the FTD may include the motion system, visual system, sound system, instructor operating station, and the normal functions and simulated malfunctions of the aircraft systems. This examination is normally accomplished simultaneously with the subjective evaluation requirements.

Appendix E1 - Aeroplane Non-Type-Specific FTD – General Simulator Requirements

- (1) This Appendix describes the general simulator requirements for qualifying an aeroplane non-type-specific FTD (Levels 1 and 2) and additional FTD qualifications for IPC and MCC. The sponsor should also consult the objective tests in Appendix E2 and functions and subjective tests listed in Appendix E3 of this to determine the complete requirements for an aeroplane specific level simulator.
- (2) The material contained in this Appendix is divided into the following categories:
 - (a) General flight deck configuration.
 - (b) Simulator programming.
 - (c) Equipment operation.
 - (d) Equipment and facilities for instructor/evaluator functions.
 - (e) Motion system.
 - (f) Visual system.
 - (g) Sound system.
- (3) Table E1A provides the standards for the General Simulator Requirements.
- (4) Table E1B provides the tasks that the sponsor will examine to determine whether the FTD satisfactorily meets the requirements for flight crew training, testing, and experience, and provides the tasks for which the simulator may be qualified.

Table E1A – Aeroplane Non-Type-Specific FTD- Minimum Simulator Requirements

Entry No.	Minimum FTD Requirements	FTD Level		Addition Qualification		Notes
		1	2	MCC	IPC	
1. General Cockpit/ Flight Deck Configuration						
1.a	Cockpit/ Flight Deck Layout And Structure					

1.a.1	An open, enclosed or perceived to be enclosed cockpit/flight deck area, excluding distraction, with aeroplane-like primary and secondary flight controls; engine and propeller controls as applicable; equipment; systems; instruments; and associated controls, assembled in a spatial manner to resemble that of the aeroplane or class of aeroplanes being simulated. The physical arrangement, appearance, and operation of controls, instruments, and switches should model at least one aircraft in the class of aircraft represented as closely as practicable. A generic aircraft instrument panel must include at least the following: Master/battery; Magnetos for each engine (as applicable); Alternators for each engine; Fuel boost pumps for each engine; Avionics master; Pitot heat; and Rotating beacon/strobe, navigation, taxi, and landing lights. The flight instrument panel(s) position and crew member seats should provide the crew member(s) with a representative posture at the controls and a representative design eye position.	X				The assembled components should be compatible and function in a cohesive manner.
1.a.2	An enclosed or perceived to be enclosed cockpit/flight deck, excluding distraction, which is spatially representative of a single set of aeroplanes including representative: primary and secondary flight controls; engine and propeller controls as applicable; systems and controls; circuit breakers; flight instruments; navigation and communications equipment; and caution and warning systems.		X	X	X	FSTD instruments and/or instrument panels using electronically displayed images with physical overlay or masking and operable controls representative of those in the aeroplane are acceptable. The instruments

Table E1A – Aeroplane Non-Type-Specific FTD- Minimum Simulator Requirements

Entry No.	Minimum FTD Requirements	FTD Level		Addition Qualification		Notes
		1	2	MCC	IPC	
	The cockpit/flight deck enclosure need only be representative of that in the single set of aeroplanes and should include windows. The enclosure need only extend to the aft end of the cockpit/flight deck. If the FSTD is used for VFR training, it should be: Fitted with a representation of a glare shield that provides the crew member(s) with a representative design eye position comparable to that of the actual aeroplane used for training. a representation of the set of aeroplanes or class of aeroplanes comparable to the actual aeroplane used for flight training.					displayed should be free of quantization (stepping). With the requirement for only a spatially representative cockpit/flight deck, the physical dimensions of the enclosure may be acceptable to simulate more than one aeroplane or class of aeroplanes in a convertible FSTD. Each FSTD conversion should be representative of single set of aeroplanes, which may require some controls, instruments, panels, masking, etc. to be changed for some conversions.
1.a.3	Cockpit/flight deck configuration must be representative of a multi-engine aeroplane. Flight controls; engine and propeller controls as applicable; systems and controls; circuit breakers; flight instruments; navigation and communications equipment; and caution and warning systems must support multi-pilot operation.			X		MCC and IPC training or checking on FTDs simulating aeroplanes that require a type-rating may only be conducted on type-specific FTDs.
1.a.4	Cockpit/flight deck must represent an aeroplane or set of aeroplanes that allow single-pilot IFR flight for Group 1 or Group 3 Instrument proficiency checking.				X	MCC and IPC training or checking on FTDs simulating aeroplanes that

Table E1A – Aeroplane Non-Type-Specific FTD- Minimum Simulator Requirements

Entry No.	Minimum FTD Requirements	FTD Level		Addition Qualification		Notes
		1	2	MCC	IPC	
	The following instruments shall be located approximately in the same positions on the instrument panels of both pilots: Airspeed indicator Attitude indicator Altimeter HIS Vertical speed indicator ADF or RMI					require a type-rating may only be conducted on type-specific FTDs.
1.b	Flight Crew Seating					
1.b.1	Flight crew member seats should provide the crew member(s) with a representative design eye position and have sufficient adjustment to allow the occupant to achieve proper posture at the controls as appropriate for the aeroplane or class of aeroplanes.	X				If a visual is installed.
1.b.2	Crew seats shall have sufficient adjustments to allow the occupant to achieve the design eye reference position appropriate to the aeroplane and for the visual system to be installed to align with that eye position.		X	X	X	

Table E1A – Aeroplane Non-Type-Specific FTD- Minimum Simulator Requirements

Entry No.	Minimum FTD Requirements	FTD Level		Addition Qualification		Notes
		1	2	MCC	IPC	
1.c	Cockpit/Flight Deck Lighting - Lighting environment for panels and instruments should be sufficient for the operation being conducted.	X	X	X	X	
2. Programming						
2.a	Aerodynamic and engine modelling must be derived from and appropriate to: An aeroplane class to support the approved use (i.e. - not required to be specific to, model, type or variant).	X		X		
	An aeroplane type or set of aeroplanes to support the approved use.		X		X	
	Flight dynamics model that accounts for various combinations of drag and thrust normally encountered in flight and supported by aeroplane generic data, including the effect of change in aeroplane attitude, sideslip, thrust, drag, altitude, temperature, gross weight, moments of inertia, centre of gravity location, and configuration.	X	X	X	X	

Table E1A – Aeroplane Non-Type-Specific FTD- Minimum Simulator Requirements

Entry No.	Minimum FTD Requirements	FTD Level		Addition Qualification		Notes
		1	2	MCC	IPC	
	As a minimum, the aeroplane's flight modeling should be comparable to a designated aeroplane class in the following areas: Aircraft performance parameters (such as maximum speed, cruise speed, stall speed, maximum climb rate) Aircraft vertical lift component must change as a function of bank modelling must reflect the rolling movement due to yawing Changes in flap setting, slat setting, gear position must be accompanied by changes in flight dynamics, The presence and intensity of wind and turbulence must be reflected in the handling and performance qualities of the simulated aircraft	X	X	X	X	
	Aerodynamic modelling must reflect the effects of airframe icing. (weight increase).			X		
2.b	Computer Capacity: Sufficient FSTD computer capacity, accuracy, resolution and dynamic response to fully support the overall FSTD fidelity needed to support the approved use.	X	X	X	X	
3. Ground Reaction and Handling Characteristics						
3.a	Generic ground reaction and ground handling models to enable touchdown effects to be reflected by the sound and visual systems.	X				Applicable if a visual is installed.

Table E1A – Aeroplane Non-Type-Specific FTD- Minimum Simulator Requirements

Entry No.	Minimum FTD Requirements	FTD Level		Addition Qualification		Notes
		1	2	MCC	IPC	
3.b	Representative aeroplane ground handling simulation to include: Ground reaction. Reaction of the aeroplane upon contact with the runway during take-off, landing and ground operations to include tire friction, side forces and other appropriate data, such as weight and speed, necessary to identify the flight condition and configuration; and Ground handling characteristics. Steering inputs to include crosswind, braking, thrust reversing, deceleration and turning radius.		X	X	X	
3.c	Runway Conditions. Stopping and directional control forces for dry runway conditions.	X	X	X	X	For Level 1: Applicable if a visual is installed. If Runway contamination effects are installed, they will be subject to evaluation.

Table E1A – Aeroplane Non-Type-Specific FTD- Minimum Simulator Requirements						
Entry No.	Minimum FTD Requirements	FTD Level		Addition Qualification		Notes
		1	2	MCC	IPC	
4. Aeroplane Systems						

Table E1A – Aeroplane Non-Type-Specific FTD- Minimum Simulator Requirements

Entry No.	Minimum FTD Requirements	FTD Level		Addition Qualification		Notes
		1	2	MCC	IPC	
4.a	Aeroplane systems installed in the FSTD should represent aeroplane system operation including system interdependencies, both on the ground and in flight. Systems should be operative to the extent that appropriate normal, abnormal and emergency operating procedures can be accomplished.	X	X	X	X	Aeroplane system operation should be predicated on, and traceable to, the system data supplied by either the aeroplane manufacturer, original equipment. Manufacturer or alternative approved data for the aeroplane system or component. Once activated, proper systems operation should result from system management by the crew member and not require any further input from the instructor's controls.
	The FSTD must include at least three (3) (designated) aircraft systems that are sufficiently operational to support multi-crew coordination training in both normal and abnormal operations. Installed systems must simulate the applicable aeroplane system operation. Systems shall be operative to the extent that it shall be possible to perform all normal, abnormal and emergency operations as may be appropriate for the aeroplane and as required for training.			X		Examples of aircraft systems that support MCC training are: pressurization systems; hydraulic systems; fuel systems; electrical system; flight control systems. The system abnormal operation should engage multi-crew activity and support MCC procedural training objectives. Sponsor must designate a minimum of 3 systems intended for MCC qualification prior to the initial evaluation.

Table E1A – Aeroplane Non-Type-Specific FTD- Minimum Simulator Requirements

Entry No.	Minimum FTD Requirements	FTD Level		Addition Qualification		Notes
		1	2	MCC	IPC	
4.b	Circuit breakers					
	A representative circuit breaker panel(s) should be presented (photographic reproductions are acceptable) and located in a spatially representative location(s).		X	X	X	
	Only those circuit breakers used in a normal, abnormal or emergency procedure need to be simulated, in a class representative form, and be functionally accurate.	X	X	X	X	
4.f	Anti-icing systems corresponding to those typically installed in that class of aeroplanes should be operative.	X	X	X	X	Simplified airframe and engine, including engine induction and pitot-static system, icing models with corresponding performance degradations due to icing should be provided. Effects of anti-icing/de-icing systems activation should also be present.

Table E1A – Aeroplane Non-Type-Specific FTD- Minimum Simulator Requirements

Entry No.	Minimum FTD Requirements	FTD Level		Addition Qualification		Notes
		1	2	MCC	IPC	
5. Physical Flight and Aircraft System Controls						
5.a	Physical flight and aircraft system controls must be recognizable as to their function and how they are to be manipulated solely from their appearance. The technique, effort, travel and direction required to manipulate physical flight and aircraft system controls must, as applicable, be representative of those in the aeroplane or class of aeroplanes Control systems should allow basic aeroplane operation with appropriate cockpit indications. Active force feedback is not required.	X				Physical flight and aircraft system controls may not be simulated using interfaces such as a keyboard, mouse, or gaming joystick to control the represented aircraft model in simulated flight. Rudder pedals must be secured to the cockpit floor structure, or physically secured to the floor beneath the device in proper relation to cockpit orientation.
5.b	Control systems operation should replicate at least one aeroplane aircraft in the set of aeroplanes simulated for normal and any non-normal modes*, including back-up systems and should reflect failures of associated systems. Appropriate cockpit indications and messages should be replicated. Active force feedback required.		X	X	X	The use of flat display technologies with physical overlays incorporating operational switches/knobs/ buttons replicating an aeroplane instrument panel may be acceptable. If the FSTD is convertible, some primary and secondary flight controls may need to be changed for some conversions. *As applicable to abnormal modes installed.

Table E1A – Aeroplane Non-Type-Specific FTD- Minimum Simulator Requirements

Entry No.	Minimum FTD Requirements	FTD Level		Addition Qualification		Notes
		1	2	MCC	IPC	
5.c	Any additional equipment installed in the FTD and intended for use in approved training must be operated in the same manner in which it would be operated in the represented aircraft. For example, landing gear, wing flaps, cowl flaps, carburetor heat control, and mixture, propeller, and throttle controls.	X	X	X	X	
5.d	Physical flight and aircraft system controls must be provided as follows as a minimum: A self-centring displacement yoke or control stick that allows continuous adjustment of pitch and bank; Self-centring rudder pedals that allow continuous adjustment of yaw; Throttle or power control(s) that allows continuous movement from idle to full power settings; Mixture/condition, propeller, and throttle/power control(s) as applicable to the aeroplane set or class;	X	X	X	X	
	Controls for the following items, as applicable to the airplane set or class represented: Wing flaps, Pitch trim (manual or electric pitch trim) permitting indicator movement either electrically or analogue in an acceptable trim ratio;	X*	X		X	The FTD must have navigation controls, displays, and instrumentation for powered aeroplanes as set out CAR 605.18 for IFR operation. * Level 1 does not require RNP approach capability. However, if installed it must function as per design.

Table E1A – Aeroplane Non-Type-Specific FTD- Minimum Simulator Requirements

Entry No.	Minimum FTD Requirements	FTD Level		Addition Qualification		Notes
		1	2	MCC	IPC	
	Communication and navigation radios, including a RNP approach capability minimum GNSS (LNAV). Clock or timer, Gear handle (if applicable), Transponder Altimeter, Microphone with push-to-talk switches, Carburetor heat (if applicable), and Cowl flaps (if applicable).					

Table E1A – Aeroplane Non-Type-Specific FTD- Minimum Simulator Requirements

Entry No.	Minimum FTD Requirements	FTD Level		Addition Qualification		Notes
		1	2	MCC	IPC	
	An FTD used for MCC training must have the following features: Retractable landing gear Pressurization system De-icing/anti-icing systems Fire detection and suppression system Dual controls Flight director Autopilot with approach mode 2 VHF transceivers Oxygen masks with intercom system 2 VHF Nav receivers (VOR ILS DME) 1 ADF receiver RNP approach capability - minimum GNSS (LNAV). 1 transponder A stopwatch accessible from both sides or two stopwatches			X		
6. Navigation and Instrument Requirements						
6.a	Navigation environment					
6.a.1	Sufficient navigation equipment necessary to fly instrument approaches to support the approved use.	X	X	X	X	

Table E1A – Aeroplane Non-Type-Specific FTD- Minimum Simulator Requirements

Entry No.	Minimum FTD Requirements	FTD Level		Addition Qualification		Notes
		1	2	MCC	IPC	
6.a.2	Instrument Indications. All relevant instrument indications involved in aeroplane or class of aeroplanes simulated should automatically respond to control movement by a flight crew member or to atmospheric disturbance and also respond to effects resulting from icing.	X	X	X	X	Numerical values should be presented in the appropriate units.
6.a.3	Navigation Database. Navigation database sufficient to support simulated aeroplane systems for real world operations.	X	X	X	X	Includes FTD database of external environment and internal aeroplane equipment databases (i.e. FMS or GPS) as applicable, to conduct RNP approaches.
6.a.4	Minimum Airport Navigation Requirement. Complete navigation database for at least 3 designated airports with corresponding precision and non-precision approach procedures, including regular updates to support approved use.	X	X	X	X	Sponsor must designate a minimum of 3 airports supporting the navigation requirement prior to the initial evaluation.
6.a.5	Instructor Controls. Instructor controls of internal and external navigational aids.	X	X	X	X	E.G. - aeroplane ILS glideslope receiver failure compared to ground facility glideslope failure. *Provide a means to stimulate a RAIM or integrity warning alert to the pilot. (See note 3 & 4 at entry # 6.a.7
6.a.6	Navigation Aids Range. Navigation aids should be usable within range or line-of-sight without restriction, as applicable to the geographic area.	X	X	X	X	Replication of the geographic environment with its specific limitations.

Table E1A – Aeroplane Non-Type-Specific FTD- Minimum Simulator Requirements

Entry No.	Minimum FTD Requirements	FTD Level		Addition Qualification		Notes
		1	2	MCC	IPC	
6.a.7	The GPS must be able to: - display the validity period or expiry date of the database and provide a means to verify serviceability of the unit (i.e. the start-up self-test feature) to support pre-flight checks required by Item 1.C. Cockpit Checks of TP 9939 (see note 1); - Display the selected approach from the GPS navigator database; - Properly and automatically sequence from each waypoint to the next during the approach; - Provide a means for the pilot to verify that the approach mode has been activated, - When the approach mode is armed, activate the approach mode within 2 NM of the Final Approach Waypoint (FAWP), - When the approach mode is activated, appropriately change the Receiver Autonomous Integrity Monitoring (RAIM) and CDI sensitivity (see note 2); - allow the pilot to fly the entire missed approach procedure including any published holds/shuttle climbs in IMC; - Have the ability to generate and display a RAIM / Integrity warning when selected from the instructor/examiner station (see note 3 & 4).		X		X	Note 1: Outdated databases are allowed, as long as matching approach plates are available and the database is updated at least annually. The FTD operator must develop processes to track and document the updates. Note 2: The GPS shall then, beginning 2 NM prior to the FAWP, have the full scale CDI sensitivity smoothly change from ± 1 NM to ± 0.3 NM at the FAWP). Note 3: The ability to recognize and manage integrity warnings is a required element of training towards an initial IFR flight test and/or an IPC, even if the recognition and management of integrity warnings are not a required part the IPC. Should an actual integrity warning occur during an initial IFR flight test that is conducted in an aircraft, candidates are expected to manage the warning appropriately. Note 4: Some integrated GNSS systems that are WAAS capable may incorporate an Integrity warning instead of a traditional RAIM warning.

Table E1A – Aeroplane Non-Type-Specific FTD- Minimum Simulator Requirements

Entry No.	Minimum FTD Requirements	FTD Level		Addition Qualification		Notes
		1	2	MCC	IPC	
6.b	Instruments and Indicators (as appropriate and applicable to the aeroplane or class of aeroplane represented)					
6.b.1	Flight instruments in a standard configuration, represented either as separate traditional “round” flight instruments, or as an electronic primary flight instrument display with reversionary and back-up flight instruments.	X	X	X	X	
6.b.2	A sensitive altimeter with incremental markings each 20 feet or less, operable throughout the normal operating range of the aeroplane or class of aeroplane represented.	X	X	X	X	
6.b.3	A magnetic compass which displays incremental markings typical of that shown in the aeroplane or class of aeroplane represented.	X	X	X	X	
6.b.4	A heading indicator with incremental markings each 5 degrees or less, displayed on a 360 degree circle. Arc segments of less than 360 degrees may be selectively displayed if desired or required, as applicable to the aeroplane or class of aeroplane represented.	X	X	X	X	
6.b.5	An airspeed indicator with incremental markings as shown on the aeroplane or class of aeroplane represented; airspeed markings of less than 40 knots need not be displayed.	X	X	X	X	
6.b.6	A vertical speed indicator with incremental markings each 100 feet per minute (fpm) for both climb and descent, for the first 1,000 fpm of climb and descent, and at each 500 fpm climb and descent for the remainder of a minimum ± 2,000 fpm total display, or as applicable to the aeroplane or class of aeroplane being represented.	X	X	X	X	

Table E1A – Aeroplane Non-Type-Specific FTD- Minimum Simulator Requirements

Entry No.	Minimum FTD Requirements	FTD Level		Addition Qualification		Notes
		1	2	MCC	IPC	
6.b.7	A gyroscopic rate-of-turn indicator or equivalent with appropriate markings for a rate of 3 degrees per second turn for left and right turns. If a turn and bank indicator is used, the 3 degrees per second rate index must be inside of the maximum deflection of the indicator.	X	X	X	X	
6.b.8	A slip and skid indicator with coordination information displayed in the conventional skid ball format where a coordinated flight condition is indicated with the ball in the centre position. A split image triangle indication may be used if applicable to the aeroplane or class of aeroplane being represented.	X	X	X	X	
6.b.9	An attitude indicator with incremental markings each 5 degrees of pitch or less, from 20 degree pitch up to 40 degree pitch down or as applicable to the aeroplane or class of aeroplane represented. Bank angles must be identified at “wings level” and at 10, 20, 30, and 60 degrees of bank (with an optional additional identification at 45 degrees) in left and right banks.	X	X	X	X	
6.b.10	Engine instruments as applicable to the aeroplane or class of aeroplane being represented, providing markings for normal ranges and minimum and maximum limits (including a fuel quantity indicator(s) that displays the fuel remaining, either in analogue or digital format).	X	X	X	X	
6.b.11	A suction gauge or instrument pressure gauge, as applicable, with a display applicable to the aeroplane or class of aeroplane represented.	X	X	X	X	
6.b.12	A flap setting indicator that displays the current flap setting. Setting indications should be typical of that found in an actual aircraft.	X	X	X	X	

Table E1A – Aeroplane Non-Type-Specific FTD- Minimum Simulator Requirements

Entry No.	Minimum FTD Requirements	FTD Level		Addition Qualification		Notes
		1	2	MCC	IPC	
6.b.13	A pitch trim indicator with a display that shows zero trim and appropriate indices of airplane nose down and airplane nose up trim, as would be found in an aircraft.	X	X	X	X	
6.b.14	Communication radio(s) with display(s) of the radio frequency in use.	X	X	X	X	If appropriate to the MCC training being conducted, an oxygen mask microphone system must be included.
	Communication equipment (intercom and air/ground) corresponding to that installed in the replicated aeroplane including an oxygen mask microphone/communication system			X		

Table E1A – Aeroplane Non-Type-Specific FTD- Minimum Simulator Requirements

Entry No.	Minimum FTD Requirements	FTD Level		Addition Qualification		Notes
		1	2	MCC	IPC	
6.b.15	Navigation radio(s) capable of replicating both precision and non-precision instrument, including approach procedures (each aural identification as applicable). The marking should include: One-half dot or less for course/glide slope deviation (i.e., VOR, LOC, or ILS), 5 degrees or less for bearing deviation for automatic direction finder (ADF) and radio magnetic indicator (RMI), as applicable, A clock with incremental markings for each minute and second, or a timer with a display of minutes and seconds, A magnetic compass with incremental markings each 10 degrees or less; The compass must display the proper lead or lag during turns, A transponder panel that displays the current transponder setting, Communications, navigation, and caution and warning equipment corresponding to that typically installed in the set of aeroplanes or class of aeroplane (as applicable) being simulated should operate within the tolerances prescribed for the applicable airborne equipment.	X	X	X	X	For example, an ILS, non-directional radio beacon (NDB), Global Positioning System (GPS), Localizer (LOC) or Very high frequency Omnidirectional Range (VOR). Graduated markings as indicated below must be present on each indication as applicable. For RNAV/RNP instrument scaling must be appropriate to the flight operational phase (i.e. enroute, terminal, and approach)

Table E1A – Aeroplane Non-Type-Specific FTD- Minimum Simulator Requirements

Entry No.	Minimum FTD Requirements	FTD Level		Addition Qualification		Notes
		1	2	MCC	IPC	
	The FTD must have navigation controls, displays, and instrumentation for powered aeroplanes as set out CAR 605.18 for IFR operation. and permit the track of the aircraft to be monitored by means of information presented on a Primary Flight display (Horizontal Situation Indicator (HSI) or Course Deviation Indicator (CDI);				X	For IPC qualification, the FTD must include navigation controls, displays, and instrumentation and navigation malfunctions necessary to conduct an Instrument Proficiency Check as prescribed in TP9939 – Flight Test Guide as amended.
6.b.16	Displays must reflect dynamic behaviour of an actual aircraft display (e.g., a VSI reading of 500 fpm must reflect a corresponding movement in altimeter, and an increase in power must reflect an increase in rpm indication or power indication).	X	X	X	X	

6.b.17	Display update must be at a frequency of 10 hertz (Hz) or higher. Each display must sense a change and react at a value less than the stated. Display updates must display all changes (within the total range of the replicated instrument) that are equal to or greater than the values stated below: Airspeed indicator: change of 5 knots, Attitude indicator: change of 2 degrees in pitch and bank, Altimeter: change of 10 feet, Turn and bank: change of ¼ standard rate turn Heading indicator: change of 2 degrees, Vertical speed indicator (VSI): change of 100 fpm, Tachometer: change of 25 rpm or 2 percent of turbine speed, VOR/ILS: change of 1 degree for VOR or ¼ of 1 degree for ILS, ADF: change of 2 degrees, GPS: Change as appropriate for model of GPS-based navigator represented, and Clock or timer: change of 1 second	X	X	X	X	
6.b.18	All instrument displays listed above must be visible during all flight operations. The update rate of all displays must provide an image of the instrument that: Does not appear to be out of focus or illegible, Does not appear to “jump” or “step” to a distracting degree during	X	X	X	X	

Table E1A – Aeroplane Non-Type-Specific FTD- Minimum Simulator Requirements

Entry No.	Minimum FTD Requirements	FTD Level		Addition Qualification		Notes
		1	2	MCC	IPC	
	operation, and Does not appear with distracting jagged lines or edges.					
	Primary flight and navigation instruments approximately life sized that exhibit neither stepping nor excessive transport delay, and arranged so as to observe trends and provide a realistic scan pattern; Digital avionics panel as applicable to approved use; GPS navigator		X	X	X	
7. Visual System Note: Level 1 FTD does not require a visual system; however, if one is installed it must meet the minimum requirements for a Level 2 FTD as indicated in this section (7).						
7.a.	If a visual system is installed, it must meet the following criteria:					
7.a.1	The visual system must respond to abrupt input at the pilot's position. An SOC is required.		X	X	X	
7.a.2	The visual system must be at least a single channel, non-collimated display. An SOC is required.		X			

Table E1A – Aeroplane Non-Type-Specific FTD- Minimum Simulator Requirements

Entry No.	Minimum FTD Requirements	FTD Level		Addition Qualification		Notes
		1	2	MCC	IPC	
7.a.3	A field of view of a minimum of 45° horizontally and 30° vertically for each training pilot position, unless restricted by the type of aeroplane. An SOC is required.		X			For flat-panel (direct view) displays: The FOV must be continuous and free of physical obstructions or viewing plane discontinuities (i.e. Panel joints) as measured from the training pilot's eye-point. The visual horizon must be continuously aligned across all adjoining panels and with and respond in a synchronized manner to pilot inputs.
7.a.4	Continuous visual field of view providing each pilot with minimum of 150° horizontal and 30° vertical field of view, unless restricted by the type of aeroplane designated. The visual system must be installed to align with that design eye reference position.		X	X	X	*Required for Level 2 if VFR cross-country training is intended.
7.a.5	The visual system must provide for a maximum parallax of 10° per pilot. An SOC is required.			X		
7.a.6	Non-collimated or direct view visual system must have the capability to align the visual presentation to the pilot flying.		X	X	X	

Table E1A – Aeroplane Non-Type-Specific FTD- Minimum Simulator Requirements

Entry No.	Minimum FTD Requirements	FTD Level		Addition Qualification		Notes
		1	2	MCC	IPC	
7.a.7	The visual scene content may not be distracting. An SOC is required.		X	X	X	
7.a.8	The minimum distance from the pilot's eye position to the surface of a direct view display may not be less than the distance to any front panel instrument. An SOC is required.		X	X	X	For flat-panel (direct view) displays, the minimum distance of side panels must be no closer to the pilot eye-point than that forward panels.
7.a.9	The visual system must provide for a minimum resolution of 5 arc-minutes for both computed and displayed pixel size. An SOC is required.		X	X		
7.a.10	Display resolution is adequate to support the approved use.		X			
7.a.11	Light-Point Size Light-point size — not greater than 8 arc minutes.		X	X	X	
7.a.12	Display Contrast Ratio Suitable to support the approved use.		X	X	X	
7.a.13	Light-Point Contrast Ratio Suitable to support the approved use.		X	X	X	
7.a.14	Light-Point Brightness Light-point brightness – not less than 20 cd/m ² (5.8 foot-lamberts).		X	X	X	

Table E1A – Aeroplane Non-Type-Specific FTD- Minimum Simulator Requirements

Entry No.	Minimum FTD Requirements	FTD Level		Addition Qualification		Notes
		1	2	MCC	IPC	
7.a.15	Display Brightness Display brightness should be demonstrated using a raster drawn test pattern. The surface brightness should not be less than 14 cd/m2 (4.1 foot-lamberts).		X	X	X	
7.a.16	Black Level and Sequential Contrast (Light valve systems only) Suitable to support the approved use.		X	X	X	
7.a.17	Motion Blur (Light valve systems only) Suitable to support the approved use.		X	X	X	
7.a.18	Speckle Test (Laser systems only) Suitable to support the approved use.		X	X	X	
7.a.19	The FTD must provide visual effects for: Light poles; Raised edge lights as appropriate; and Glow associated with approach lights in low visibility before physical lights are seen (required only if low visibility training is intended.)				X	Visual effects for light poles and raised edge lights are for the purpose of providing additional depth perception during takeoff, landing, and taxi training tasks. Three dimensional modeling of the actual poles and stanchions is not required.
8. External Environment						

Table E1A – Aeroplane Non-Type-Specific FTD- Minimum Simulator Requirements

Entry No.	Minimum FTD Requirements	FTD Level		Addition Qualification		Notes
		1	2	MCC	IPC	
8.a	Atmosphere And Weather					
8.a.1	Simulation of the standard atmosphere including instructor control over key parameters.	X	X	X	X	
8.a.2	The FSTD should employ wind shear models that provide training for recognition of wind shear phenomena.		X	X		
8.a.3	Visibility as observed on the visual system should be simulated and respective instructor controls provided.		X	X	X	
8.b	Airports And Terrain					
8.b.1	The system should provide a visual scene with sufficient content to allow a pilot to successfully accomplish a visual landing. Scenes should include a definable horizon and typical terrain characteristics such as fields, roads and bodies of water and surfaces illuminated by aeroplane landing lights to support the approved use.		X	X	X	For specific VFR cross-country training the capability to replicate ground visual references and topographical features sufficient to support VFR navigation according to appropriate charts — minimum standard 1:500 000 scale mapping.
	Correct terrain modelling, runway orientation, markings, lighting, dimensions and taxiways to support approved use.					
	Generic airport models with topographical features to support the approved use.	X				If visual installed

Table E1A – Aeroplane Non-Type-Specific FTD- Minimum Simulator Requirements

Entry No.	Minimum FTD Requirements	FTD Level		Addition Qualification		Notes
		1	2	MCC	IPC	
	Minimum of one (1) designated airport-specific model with sufficient scene content to allow a pilot to successfully accomplish a visual landing. Scenes should include a definable horizon and typical terrain characteristics such as fields, roads and bodies of water and surfaces illuminated by aeroplane landing lights. Visual terrain and EGPWS databases should be matched to support procedural CFIT training if intended.		X	X		Where the device is required to perform low visibility operations, at least one (1) specific airport scene with functionality to support the approved use. (e.g. low visibility taxi route with marker boards, stop bars, runway guard lights plus the required approach and runway lighting.) Sponsor must designate a minimum airport specific models prior to the initial evaluation.
	Minimum of three (3) designated airport-specific models with sufficient scene content to allow a pilot to successfully accomplish a visual landing. Scenes should include a definable horizon and typical terrain characteristics such as fields, roads and bodies of water and surfaces illuminated by aeroplane landing lights to support the approved use. Each designated airport must have at least one Required Navigation Performance (RNP) approach with all waypoints as named on the approach plate in the correct order and with the correct bearing and distances between waypoints is available at each airport;				X	
8.c	Visual Cues					
8.c.1	This should include: Surface on runways, taxiways, and ramps; Terrain features;		X	X	X	

Table E1A – Aeroplane Non-Type-Specific FTD- Minimum Simulator Requirements

Entry No.	Minimum FTD Requirements	FTD Level		Addition Qualification		Notes
		1	2	MCC	IPC	
	For specific VFR cross-country training, highly detailed and accurate surface depiction of the terrain surface within an area sufficient to achieve cross-country flying under VFR conditions.		X			
	Visual cues to assess sink rate and depth perception during take-off and landing should be provided.		X	X	X	
	Highly detailed and accurate surface depiction of the terrain surface within an approximate area from 400 m (1/4 sm) before the runway approach end to 400 m (1/4 sm) beyond the runway departure end with a total width of approximately 400 m (1/4 sm) including the width of the runway. When the aircraft reaches the Missed Approach Point (MAP), provide visuals that correspond to the position of the aircraft and provide sufficient realism to allow the candidate to make the decision to continue the approach with a normal visual transition and landing or initiate the missed approach procedure.				X	
8.d	Attitude					
8.d.1	The FSTD should provide for accurate portrayal of the visual environment relating to the FSTD attitude.	X	X	X	X	For Level 1 – if visual installed.
8.e.	Airport Scenes					

Table E1A – Aeroplane Non-Type-Specific FTD- Minimum Simulator Requirements

Entry No.	Minimum FTD Requirements	FTD Level		Addition Qualification		Notes
		1	2	MCC	IPC	
8.e.1	The system should include at least 3 designated real-world airports available in daylight, twilight (dusk or dawn) and night illumination states.				X	
8.e.2	The system should include at least 1 designated real-world airport available in daylight, twilight (dusk or dawn) and night illumination states.		X	X		
8.e.4	Daylight Capability.		X	X	X	
8.e.5	The system should provide full-colour presentations and sufficient surfaces with appropriate textural cues to successfully accomplish a visual approach, landing and airport movement (taxi).					
8.e.6	Surface shading effects should be consistent with simulated sun position.		X	X	X	
8.e.7	Total scene content comparable in detail to that produced by 10 000 visible textured surfaces and 6 000 visible lights should be provided.		X	X	X	
8.e.8	Total scene content should be sufficient to identify the airport and represent the surrounding terrain.		X	X	X	
8.e.9	The system should have sufficient capacity to display 16 simultaneously moving objects.		X	X	X	
8.e.10	Twilight (dusk) capability.		X	X	X	

Table E1A – Aeroplane Non-Type-Specific FTD- Minimum Simulator Requirements

Entry No.	Minimum FTD Requirements	FTD Level		Addition Qualification		Notes
		1	2	MCC	IPC	
8.e.11	The system should provide twilight (or dusk) visual scenes with full colour presentations of reduced ambient intensity and typical terrain characteristics such as fields, roads and bodies of water and surfaces illuminated by representative own-ship-lighting (e.g. landing lights) sufficient to successfully accomplish visual approach, landing and airport movement (taxi).		X	X	X	
8.e.12	Total scene content comparable in detail to that produced by 10 000 visible textured surfaces and 15 000 visible lights should be provided.		X	X	X	
8.e.13	Scenes should include self-illuminated objects such as road networks, ramp lighting and airport signage, to conduct a visual approach, landing and airport movement (taxi).		X	X	X	
8.e.14	The system should include a definable horizon.		X	X	X	
8.e.15	The system should have sufficient capacity to display 16 simultaneously moving objects.		X	X	X	
8.e.16	Night capability.		X	X	X	
8.e.17	The system should provide at night all features applicable to the twilight scene, as defined above, with the addition of the need to portray reduced ambient intensity that removes ground cues that are not self-illuminating or illuminated by aeroplane lights (e.g. landing lights).		X	X	X	

Table E1A – Aeroplane Non-Type-Specific FTD- Minimum Simulator Requirements						
Entry No.	Minimum FTD Requirements	FTD Level		Addition Qualification		Notes
		1	2	MCC	IPC	
8.e.18	Airport Clutter. Airport models should include representative static and dynamic clutter such as gates, aeroplanes, and ground handling equipment.		X	X	X	
9. Motion system.						
9.a.	The FTD may have a motion system, if desired, although it is not required. If a motion system is installed, the motion system operation must not be distracting and must be coupled closely to provide integrated sensory cues. The motion system must also respond to abrupt input at the pilot's position within the allotted time, but not before the time when the aeroplane responds under the same conditions.	X	X			The motion system standards set out in this manual, Appendix A1 for at least Level A simulators are acceptable.
10. System integration						
10.a	Transport delay: Transport delay may not exceed 300 milliseconds. An SOC is acceptable in lieu of testing, however, testing for initial evaluation may be required to verify the SOC if latency calculations in the SOC are close to tolerance.			X	X	If a motion system is installed, it must be measured by latency tests or transport delay tests and may not exceed 300 milliseconds. Instrument and visual response may not occur prior to motion onset.
11. Sound system						

Table E1A – Aeroplane Non-Type-Specific FTD- Minimum Simulator Requirements

Entry No.	Minimum FTD Requirements	FTD Level		Addition Qualification		Notes
		1	2	MCC	IPC	
11.a	Significant sounds perceptible to the flight crew during flight operations to support the approved use. Comparable engine and airframe sounds.	X				
11.b	Significant cockpit/flight deck sounds during normal and abnormal operations, aeroplane class-like, including engine and airframe sounds as well as those which result from pilot or instructor-induced actions.		X	X	X	
11.c	The volume control must have an indication of sound level setting for approved use.		X	X	X	This indication is of the sound level setting as evaluated during the FTD's initial evaluation.
11.d	The sound of a crash when the simulated aeroplane exceeds limitations.		X	X	X	
11.e	The volume control should have an indication of sound level setting for approved use. Full volume should correspond to actual volume level agreed at the initial evaluation. When full volume is not selected, an indication of abnormal setting should be provided to the instructor.	X	X	X	X	

Table E1A – Aeroplane Non-Type-Specific FTD- Minimum Simulator Requirements						
Entry No.	Minimum FTD Requirements	FTD Level		Addition Qualification		Notes
		1	2	MCC	IPC	
12. Instructor or Evaluator Facilities.						

Table E1A – Aeroplane Non-Type-Specific FTD- Minimum Simulator Requirements

Entry No.	Minimum FTD Requirements	FTD Level		Addition Qualification		Notes
		1	2	MCC	IPC	
12.a	In addition to the flight crew member seats, there should be an instructor station seat and one suitable seats for an observer or a TC inspector. The instructor station should provide an adequate view of the pilots' panels and forward windows.	X	X	X	X	For MCC and IPC qualifications - At least one seat should have a system to permit monitoring of all flight crew member and instructor communications. These seats need not be a replica of an aircraft seat and may be as simple as an office chair placed in an appropriate position. The MSP will consider alternatives alternate-seating based on unique flight deck configurations. For an FSTD with a motion cueing system, any instructor seat on board the motion platform must be adequately secured and fitted with positive restraint devices of sufficient integrity to safely restrain the occupant during any known or predicted motion system excursion.

Table E1A – Aeroplane Non-Type-Specific FTD- Minimum Simulator Requirements

Entry No.	Minimum FTD Requirements	FTD Level		Addition Qualification		Notes
		1	2	MCC	IPC	
12.b	The FTD must have instructor controls that permit: Control of all required system variables, freezes, resets, and Activation of failures in navigation and instruments, radio receivers, landing gear and flaps, engine power (partial and total), and other aircraft systems (pitot, electric, static, etc.) to support the approved use. Once activated, proper system operation must result from system management by the crew and not require input from the instructor controls	X	X	X	X	The effects of malfunctions should be sufficient to correctly exercise the procedures in relevant operating manuals.
12.c	The FTD must have IOS instructor controls for the following expected environmental effects to support the approved use:					
	Surface wind speed, direction and gusts	X	X	X	X	
	Intermediate and high altitude wind speed and direction as appropriate;			X	X	
	Thunderstorms and microbursts; and			X		
	Turbulence.					Only instrument effects are required.
12.d	The simulator must provide the instructor or evaluator the ability to present ground and air hazards.			X	X	For example, another aeroplane crossing the active runway or converging airborne traffic.

Table E1A – Aeroplane Non-Type-Specific FTD- Minimum Simulator Requirements

Entry No.	Minimum FTD Requirements	FTD Level		Addition Qualification		Notes
		1	2	MCC	IPC	
12.e	Oversee tracks along airways, holding entries and patterns, and localizer and glide slope alignment/deviation (or other approaches with a horizontal and vertical track);		X		X	

Note: Items not installed or not functional on the FTD and not appearing on the Qualification Certificate Configuration List, are not required to be listed as exceptions on the Qualification Certificate.

Table E1B – Aeroplane Non-Type-Specific FTD - Subjective Requirements					
Entry #	Functions and Subjective Tests	Level		Qualification	
		1	2	- MCC	-IPC
1- 2 Preflight Procedures.					
1.	Pre-flight Inspection (flight deck only) (Appropriate checklists must be available in FTD).	X	X	X	X
2.a	Engine Start	X	X	X	X
2.b	Taxiing		X	X	X
3. Takeoff and Departure Phase.					
3.a.1	Pre-takeoff Checks	X	X	X	X
3.a.2	Normal and Crosswind Takeoff		X	X	X
3.a.3	Instrument Takeoff		X	X	X
3.b	Rejected Takeoff - Consistentlycontrollable within runwaywidth to support approved use.		X	X	X
3.c	Departure Procedure		X	X	X
4-6 Inflight Maneuvers.					
4.	Normal climb	X	X	X	X
5.a	Steep Turns	X	X	X	X
5.b	Approaches to Stall	X	X	X	X
5.c	Engine Failure—Multiengine Aeroplane		X	X	X
5.d	Engine Failure—Single-Engine Aeroplane		X		
5.e	Specific Flight Characteristics incorporated for TC approved flight training program.	X	X	X	X
6.	Normal decent	X	X	X	X
7. Instrument Procedures					
7.a	Standard Terminal Arrival / Flight Management System Arrivals Procedures	X	X	X	X
7.b	Holding	X	X	X	X
7.c	Precision Instrument - All engines operating.	X	X	X	X
7.d	Precision Instrument - One engine inoperative.	X	X	X	X
7.e	Non-precision Instrument Approach.(For IPC, the FTD must simulate an RNP approach.)	X	X	X	X
8. Landings and Approaches to Landings.					
8.a	Normal and Crosswind Visual Approaches and Landings		X	X	X
8.b	Landing From a Precision / Non-Precision Approach		X	X	X
8.c	Approach and Landing with (Simulated) Engine Failure – Multiengine Aeroplane		X	X	X
8.e	Rejected Landing		X	X	X
9. Missed Approach.					
9.a	Normal.	X	X	X	X
9.b	One engine Inoperative.	X	X	X	X
10. Post-Flight Procedures.					
10.a	After-Landing Procedures	X	X	X	X

Table E1B – Aeroplane Non-Type-Specific FTD - Subjective Requirements					
Entry #	Functions and Subjective Tests	Level		Qualification	
		1	2	- MCC	-IPC
10.b	Parking and Securing	X	X	X	X
11. Normal and Abnormal Procedures. X* - indicates that the system, task, or procedure, although not required to be present, may be examined if the appropriate aeroplane system is simulated in the FTD and is working properly.					
11.a	Engine (including shutdown and restart)	X	X	X	X
11.b	Fuel System	X	X	X	X
11.c	Electrical System	X	X	X	X
11.d	Hydraulic System		X*	X*	X*
11.e	Environmental and Pressurization Systems		X*	X*	X*
11.f	Fire Detection and Extinguisher Systems		X*	X*	X*
11.g	Navigation and Avionics Systems	X	X	X	X
11.h	Flight Control Systems	X	X	X	X
11.i	Anti-ice and Deice Systems	X	X	X	X
11.j	Aircraft and Personal Emergency Equipment		X*	X*	
11.k	Equipped and serviceable for instrument flight as set out in CAR 605.18 for IFR operations (as amended): attitude indicator; vertical speed indicator; outside air temperature gauge; anti-icing for each airspeed indicating system; a power failure warning device or vacuum indicator showing power available to gyroscopic instruments from each power source; an alternative source of static pressure for the altimeter, airspeed indicator and vertical speed indicator; sufficient radio communication equipment; sufficient radio navigation equipment to permit the pilot, in the event of the failure at any stage of the flight of any item of that equipment, including any associated flight instrument display, where the aircraft is operated in IMC, to complete an instrument approach and, if necessary, conduct a missed approach procedure to proceed to the destination aerodrome or proceed to another aerodrome that is suitable for landing. (Night) a means of illumination for all instruments used to operate the aircraft; (Night) a landing light;				X
11.l	Emergency Procedures. Emergency Descent (Max. Rate) Inflight Fire and Smoke Removal Rapid Decompression Emergency Evacuation		X*	X*	
12. Visual System (Appropriate for approved use)			X	X	X
16. Instructor Operating Station (ios) functions, as appropriate.					
16.a	Power switch (es).	X	X	X	X

Table E1B – Aeroplane Non-Type-Specific FTD - Subjective Requirements

Entry #	Functions and Subjective Tests	Level		Qualification	
		1	2	- MCC	-IPC
16.b	Aeroplane conditions	X	X	X	X
16.c	Airports / Runway.	X	X	X	X
16.d	Environmental controls	X	X	X	X
16.e	Aeroplane system malfunctions (insertion / deletion)	X	X	X	X
16.f	Locks, Freezes, and Repositioning	X	X	X	X
16.g	Sound Controls - On / Off / adjustment	X	X	X	X
16.h	Motion / Control Loading System - On / off / emergency stop As appropriate.	X	X	X	X
16.i	Observer Seats / Stations: Position / Adjustment / restraint (as appropriate).	X	X	X	X

Appendix E2 - Aeroplane Non-Type-Specific FTD – Objective Tests

- (1) In addition to subjective testing, non-type-specific FTDs will be subject to the applicable objective tests at Table E2A to ensure they meet qualification criteria initially and periodically throughout its life. The objective tests can be flown by a suitably skilled person and the results recorded manually.
- (2) The tolerances specified are designed to ensure that the device meets its original target criteria year after year. It is therefore important that such target data is carefully derived and substantiated prior to submitting the QTG to the inspecting authority prior to the QTG review and initial qualification. For initial qualification, it is highly desirable that the device should meet its design criteria within the listed tolerances. However, unlike the tolerances stipulated for type-specific devices, the tolerances stated for non-type-specific FSTDs are intended to support recurrent inspection to ensure the device is maintained as approved during the initial evaluation. It is therefore important that the test targets be representative and test results be repeatable during the life of the FSTD.
- (3) A number of tests within the QTG have had their tolerances reduced to "Correct Trend and Magnitude" (CT&M) thereby avoiding the need for specific validation data. The use of CT&M is not to be taken as an indication that certain areas of simulation can be ignored. For such tests, the performance of the FSTD should be appropriate and representative of the simulated designated aeroplane and should under no circumstances exhibit negative characteristics. Where CT&M is used as a tolerance, it is strongly recommended that targets are identified and used to footprint the baseline in order to avoid possible divergent subjective opinions during recurrent evaluations. Once a footprint target is established, then the applicable tolerance should be applied to results during recurrent evaluations.
- (4) Longitudinal change force tests. It is acceptable to use change force tests instead of the longitudinal power, flap and gear change dynamics tests. The purpose of these change force tests is to simply reproduce the control force feeling during recurrent evaluations. If not otherwise stated, these tests should be conducted in the following way:
 - (a) trim the aeroplane in straight and level flight in the most suitable configuration;
 - (b) initiate the configuration change (power, flap or gear change);
 - (c) maintain the airspeed constant using the pitch control (or as an alternate procedure, maintain the altitude of the original trimmed conditions); and
 - (d) Measure the required pitch control force change.

Note: If change forces are small (within the tolerance band), it is acceptable to reduce change the end state airspeed to achieve a more measureable force provided the procedure is well documented and repeatable.

Table E2A – Aeroplane – Non- Type-Specific FTD - Objective Tests

Test		Tolerance	Flight Conditions	Test Details	Level		Qualification		Notes
Entry #	Title				1	2	MCC	IPC	
1. Performance									
1.b	Ground acceleration time and distance.	±1.5 s or ±5% of time.	Take-off.	Acceleration time and distance should be recorded for a minimum of 80% of the total time from brake release to Vr. May be combined with normal take-off (1.b.4) or rejected take-off (1.b.7).		X			Plotted data should be shown using appropriate scales for each portion of the manoeuvre
	Rejected take-off.	±5% of time or ±1.5 s.	Take-off.	Record time for at least 80% of the time segment from initiation of the rejected takeoff to full stop.					
1.c.1	Normal Climb, all engines operating.	±3 kt airspeed. ±0.5 m/s (100 ft/ min) or ±5% of rate of climb.	Clean	Flight test data are preferred; however, aeroplane performance manual data are an acceptable alternative. Record at nominal climb speed and mid initial climb altitude. FTD performance is to be recorded over an interval of at least 300 m (1 000 ft).	T & M	X			This may be a snapshot test result.

Table E2A – Aeroplane – Non- Type-Specific FTD - Objective Tests

Test		Tolerance	Flight Conditions	Test Details	Level		Qualification		Notes
Entry #	Title				1	2	MCC	IPC	
1.c	One-engine inoperative 2nd segment climb.	±3 kt airspeed. ±0.5 m/s (100 ft/ min) or ±5% of rate of climb, but not less than aeroplane performance data requirements.	2nd segment climb.			T & M			This may be a snapshot test result.
1.d.1	Level flight acceleration.	±5% of time.	Cruise.	Time required to decrease airspeed a minimum of 50 kt, using idle power. For aeroplanes with a small operating speed range, speed change may be reduced to 80% of operational speed change.		X			
1.d.2	Level flight deceleration.	±5% of time.	Cruise.	Time required to decrease airspeed a minimum of 50 kt, using idle power. For aeroplanes with a small operating speed range, speed change may be reduced to 80% of operational speed change.		X			

Table E2A – Aeroplane – Non- Type-Specific FTD - Objective Tests

Test		Tolerance	Flight Conditions	Test Details	Level		Qualification		Notes
Entry #	Title				1	2	MCC	IPC	
1.f.1	Acceleration	±10% Ti or ± 1 s; and ±10% Tt or ± 1 s.	Approach or landing	Total response is the incremental change in the critical engine parameter from idle power to go-around power.	T & M	X			Ti = total time from initial throttle movement until a critical engine parameter reaches 10% of its total response below maximum take-off power.
1.f.2	Deceleration	±10% Ti or ± 1 s; and ±10% Tt or ± 1 s..	Ground	Total response is the incremental change in the critical engine parameter from idle power to go-around power.	T & M	X			Tt = total time from initial throttle movement until a critical engine parameter reaches 90% of its total response below maximum take-off power. Total response is the incremental change in the critical engine parameter from maximum take-off power to idle power.
2.a.1	Pitch controller position versus force	±0.9 daN (2 lbf) breakout. ±2.2 daN (5 lbf) or ±10% of force.	As determined by sponsor	Control forces and travel should broadly correspond to that of the replicated class of aeroplane.	T & M				

Table E2A – Aeroplane – Non- Type-Specific FTD - Objective Tests

Test		Tolerance	Flight Conditions	Test Details	Level		Qualification		Notes
Entry #	Title				1	2	MCC	IPC	
				Record uninterrupted control sweep to stops during initial qualification The recorded tolerances apply to subsequent comparisons on continuing qualification evaluations.					Test results should be validated with in-flight data from tests such as longitudinal static stability, stalls, etc.
2.a.2	Roll controller position versus force	±0.9 daN (2 lbf) breakout. ±1.3 daN (3 lbf) or ±10% of force.	As determined by sponsor	Control forces and travel should broadly correspond to that of the replicated class of aeroplane.	T & M				
				Record uninterrupted control sweep to stops during initial qualification The recorded tolerances apply to subsequent comparisons on continuing qualification evaluations.		X			Test results should be validated with in-flight data from tests such as engine-out trims, steady state side-slips, etc.
2.a.3	Rudder pedal position versus force	±2.2 daN (5 lbf) breakout. ±2.2 daN (5 lbf) or ±10% of force.	As determined by sponsor	Control forces and travel should broadly correspond to that of the replicated class of aeroplane.	X				
				Record uninterrupted control sweep to stops during initial qualification The recorded tolerances apply to subsequent comparisons on continuing qualification evaluations.		X			Test results should be validated with in-flight data from tests such as engine-out trims, steady state side-slips, etc.

Table E2A – Aeroplane – Non- Type-Specific FTD - Objective Tests

Test		Tolerance	Flight Conditions	Test Details	Level		Qualification		Notes
Entry #	Title				1	2	MCC	IPC	
2.a.4.a	Nosewheel steering controller force and position calibration.	Correct trend and magnitude	Ground	Suitable for approved use.	T & M	T & M	T & M	T & M	As applicable to replicated aeroplane.
2.a.4.b	Rudder pedal steering calibration.	Correct trend and magnitude	Ground	Suitable for approved use.	T & M	T & M	T & M	T & M	
2.a.8	Alignment of cockpit throttle lever versus selected engine parameter.	Correct trend and magnitude. Parameter +/- 5% at: full forward full aft and 50% throttle position.	Ground					T & M	Test must include a method of measuring 50% throttle position.
2.c.1.b	Power Change Force.	±5 lb (2.2 daN) or, ±20% pitch control force.	Approach	Force tests should provide the force required to maintain constant airspeed or altitude to complete the configuration change.	T & M				Force tests should provide the force required to maintain constant airspeed or altitude to complete the configuration change.
2.c.2.b	Flap/Slat Change Force.	±5 lb (2.2 daN) or, ±20% pitch control force.	Takeoff through initial flap retraction, and approach to landing.	Force tests should provide the force required to maintain constant airspeed or altitude to complete the configuration change.	T & M				Force tests should provide the force required to maintain constant airspeed or altitude to complete the configuration change.

Table E2A – Aeroplane – Non- Type-Specific FTD - Objective Tests

Test		Tolerance	Flight Conditions	Test Details	Level		Qualification		Notes
Entry #	Title				1	2	MCC	IPC	
2.c.4.b	Gear Change Force.	±5 lb (2.2 daN) or, ±20% pitch control force.	Takeoff (retraction), and Approach (extension).	Force tests should provide the force required to maintain constant airspeed or altitude to complete the configuration change.	T & M				Force tests should provide the force required to maintain constant airspeed or altitude to complete the configuration change.
2.c.5	Longitudinal Trim.	±1° elevator angle. ±0.5° stabilizer or trim surface angle. ±1° pitch angle. ±5% of net thrust or equivalent.	Cruise, Approach, and Landing.	Pitch controller position and/or trim controller position may be used instead of elevator angle and trim control position instead of stabilizer angle.	T & M				
2.c.7	Longitudinal Static Stability.	±2.2 daN (5 lbf) or ±10% of pitch controller force. Alternative method: ±1° or ±10% of the change of elevator angle.	Approach	Data for at least two speeds above and two speeds below trim speed. The speed range must be sufficient to demonstrate stick force versus speed characteristics. This test may be a series of snapshot tests.	T & M				Force tolerance is not applicable if forces are generated solely by the use of aeroplane hardware in the FTD. Alternative method applies to aeroplanes which do not exhibit speed stability characteristics.

Table E2A – Aeroplane – Non- Type-Specific FTD - Objective Tests

Test		Tolerance	Flight Conditions	Test Details	Level		Qualification		Notes
Entry #	Title				1	2	MCC	IPC	
2.c.8.b	Stall Warning (actuation of stall warning device.)	±3 kt airspeed for stall warning	Second Segment Climb, and Approach or Landing.	The stall maneuver must be entered with thrust at or near idle power and wings level (1g). Record the stall warning signal and initial buffet if applicable.					
2.c.9.b	Phugoid Dynamics.	±10% period, Representative damping.	Cruise	The test must include whichever is less of the following: Three full cycles (six overshoots after the input is completed), or the number of cycles sufficient to determine representative damping.	T & M				
2.d.2	Response (Rate).	±2°/s or ±10% of roll rate.	Cruise, and Approach or Landing.	Test with normal roll control displacement (approximately one-third of maximum roll controller travel).	T & M				
2.d.4.c	Spiral Stability.	Correct trend and ±3° or ±10% of roll angle in 20s.	Cruise	Aeroplane data averaged from multiple tests may be used	T & M				Test for both directions. As an alternative test, show lateral control required to maintain a steady turn with a roll angle of approximately 30°.

Table E2A – Aeroplane – Non- Type-Specific FTD - Objective Tests

Test		Tolerance	Flight Conditions	Test Details	Level		Qualification		Notes
Entry #	Title				1	2	MCC	IPC	
2.d.6.b	Rudder Response.	Roll rate $\pm 2^\circ/\text{sec}$, Bank angle $\pm 3^\circ$. or $\pm 2^\circ/\text{s}$ or $\pm 10\%$ of yaw rate or $\pm 10\%$ of heading change.	Approach or Landing.	May be roll response to a given rudder deflection.	T & M				Test with stability augmentation on and off. Test with a step input at approximately 25% of full rudder pedal throw.

Table E2A – Aeroplane – Non- Type-Specific FTD - Objective Tests

Test		Tolerance	Flight Conditions	Test Details	Level		Qualification		Notes
Entry #	Title				1	2	MCC	IPC	
2.d.8	Steady State Sideslip.	For a given rudder position: ±2° roll angle; ±1° side-slip angle; ±2° or ±10% of aileron angle; and ±5° or ±10% of spoiler or equivalent roll controller position or force. For aeroplanes with reversible flight control systems: ±1.3 daN (3 lbf) or ±10% of wheel force. ±2.2 daN (5 lbf) or ±10% of rudder pedal force.	Approach or Landing.	This test may be a series of snapshot tests using at least two rudder positions (in each direction for propeller-driven aeroplanes), one of which must be near maximum allowable rudder. Roll controller position instead of aileron angle may be used.	T & M				Side-slip angle is matched only for repeatability and only on continuing recurrent evaluations.

Table E2A – Aeroplane – Non- Type-Specific FTD - Objective Tests

Test		Tolerance	Flight Conditions	Test Details	Level		Qualification		Notes
Entry #	Title				1	2	MCC	IPC	
4.a.1	Visual Display field of view.	Visual field-of-view for each pilot with a minimum of 45° horizontally and 30° vertically, unless restricted by the type of aeroplane, simultaneously for each pilot.	Not applicable	The minimum distance from the pilot's eye position to the surface of a direct view display may not be less than the distance to any front panel instrument. 30° vertical field of view may be insufficient to meet the requirements of the visual ground segment (if required). This needs to be considered in the FOV calculation.		X			For flat-panel (direct view) displays: The FOV must be continuous and free of physical obstructions or viewing plane discontinuities (i.e. Panel joints) as measured from the training pilot's eye-point. The visual horizon must be continuously aligned across all adjoining panels and with and respond in a synchronized manner to pilot inputs. 30° vertical field of view may be insufficient to meet the requirements of the visual ground segment (if required).

Table E2A – Aeroplane – Non- Type-Specific FTD - Objective Tests

Test		Tolerance	Flight Conditions	Test Details	Level		Qualification		Notes
Entry #	Title				1	2	MCC	IPC	
		Visual display providing each pilot with a minimum of 150° horizontal and 40° vertical continuous field of view.					X	X	Field of view should be measured using a visual test pattern filling the entire visual scene (all channels) consisting of a matrix of black and white 5° squares. Installed alignment should be confirmed in an SOC (this would generally consist of results from acceptance testing).

Appendix E3 – Aeroplane – Non-Type-Specific FTD –Subjective Evaluation

Table E3A – Aeroplane Non-Type-Specific FTD - Subjective Requirements

Entry #	Functions and Subjective Tests	Level		Qualification		Comment
		1	2	MCC	IPC	
1. Preparation for flight.						
1.a	Pre-flight. Note: Accomplish a functions check of all switches, indicators, systems, and equipment at all crew members' and instructors' stations and determine that:					
1.a.2	The flight deck design and functions maybe generic, but are aeroplane-like and recognizable as within a class of aeroplanes.	X		X		
1.a.3	The flight deck design and functions represent those of the simulated aeroplanes or set of aeroplanes.		X		X	
2. Surface operations (pre-flight).						
2.a	Engine start.					
2.a.1	Normal start.	X	X	X	X	
2.a.2	Alternate start procedures.		X	X		
2.a.3	Abnormal starts and shutdowns (hot start, hung start, tail pipe fire, etc.).		X	X		
2.b	Taxi.					
2.b.1	Pushback/powerback.	N/A	N/A			
2.b.2	Thrust response.	X	X	X	X	
2.b.3	Power lever friction.	X	X	X	X	
2.b.4	Ground handling.	X*	X	X	X	*If a visual is installed.
2.b.6	Low visibility (taxi route, signage, lighting, markings, etc).		X*	X	X	Applicable if low visibility training is intended. *Visual FOV must be 150x30 degrees minimum.
2.c	Brake operation:					
2.c.1	Normal, automatic and alternate/emergency operation.	X	X	X	X	
3. Take-off.						
3.a	Normal. Note: Only applicable to those take-off tests relevant to the simulated aeroplane type or class. Conduct tests with limiting wind velocities, wind shear and with relevant system failures.					
3.a.1	Aeroplane/engine parameter relationships, including run-up.	X	X	X	X	
3.a.2	Nosewheel and rudder steering. (as applicable)	X*	X	X	X	*If visual is installed.
3.a.3	Crosswind (maximum demonstrated).	X	X	X	X	If visual is installed.
3.a.5	Low visibility take-off.		X	X*	X*	Applicable if low visibility training is intended. *Visual FOV must be 150x30 degrees minimum.
3.a.6	Landing gear, wing flap and leading edge device operation.	X	X	X	X	
3.a.7	Other.		X	X	X	As applicable if intended for training

Table E3A – Aeroplane Non-Type-Specific FTD - Subjective Requirements

Entry #	Functions and Subjective Tests	Level		Qualification		Comment
		1	2	MCC	IPC	
3.b	Abnormal/emergency.					
3.b.1	Rejected take-off.			X	X	Consistently controllable within runway width to support approved use.
3.b.4	Continued take-off with failure of most critical engine at most critical point.			X		Controllable to support procedural training.
3.b.5	Flight control system failures, reconfiguration modes, manual reversion and associated handling.			X		As applicable to systems installed.
4. Climb.						
4.a	Normal.	X	X	X	X	
4.b	One or more engine(s) inoperative.		X	X	X	
5. Cruise.						
5.a	Performance characteristics (speed versus power, configuration, and attitude).					
5.a.1	Straight and level flight.	X	X	X	X	
5.a.2	Change of airspeed.	X	X	X	X	
5.a.3	High-altitude handling.			X		
5.a.5	Overspeed warning (in excess of V _{mo} or M _{mo}).			X	X	
5.a.7	Other. As applicable to intended additional approved training					
5.b	Manoeuvres.					
5.b.1	High angle of attack, approach to stalls, stall warning, buffet, and g-break (take-off, cruise, approach, and landing configuration).	X	X	X	X	
5.b.2	Slow flight.	X	X	X	X	
5.b.3	Spin.					
5.b.4	Flight envelope protection (high AOA, bank limit, overspeed, etc.).			X	X	As applicable
5.b.6	Normal and standard rate turns.	X	X	X	X	
5.b.7	Steep turns.	X	X	X	X	
5.b.8	Performance turn.	X	X	X	X	
5.b.9	In-flight engine shutdown and restart (assisted and windmill).		X	X	X	As applicable
5.b.10	Manoeuvring with one or more engines inoperative.		X	X	X	As applicable
5.b.11	Specific flight characteristics (e.g. direct lift control).			X		
5.b.12	Flight control system failures, reconfiguration modes, manual reversion and associated handling.		X	X	X	As applicable
5.b.13	Gliding to a forced landing.		X	X		

Table E3A – Aeroplane Non-Type-Specific FTD - Subjective Requirements

Entry #	Functions and Subjective Tests	Level		Qualification		Comment
		1	2	MCC	IPC	
5.b.14a	Visual resolution and FSTD handling and performance for the following: Terrain accuracy for forced landing area selection; Terrain accuracy for VFR Navigation; Eights on pylons (visual resolution); Turns about a point. S-turns about a road or section line		X			As applicable and approved for use
5.b.15	Other. As applicable to intended additional approved training		X			
6. Descent.						
6.a	Normal.	X	X	X	X	
6.b	Maximum rate/emergency(clean, with speedbrakes, etc.).	X	X	X		As applicable
6.c	With autopilot.	X	X	X	X	As applicable
6.d	Flight control system failures, reconfiguration modes, manual reversion and associated handling.			X		As applicable
6.e	Other. As applicable to intended additional approved training	X	X	X	X	
7. Instrument approaches and landing. (As applicable) Note: Only applicable to instrument approach and landing tests relevant to the simulated aeroplane set/ class or as required for approved use. Conduct tests using limiting wind velocities, wind shear (except for the CAT II and III precision approaches) and with relevant system failures.						
7.a	Precision approach.					
7.a.1	CAT I published approaches.					
7.a.1.a	– Manual approach with/without flight director including landing.	X	X	X	X	
7.a.1.b	– Autopilot coupled approach and manual landing.		X	X	X	
7.a.1.c	– Autopilot coupled approach, engine(s) inoperative.		X	X	X	
7.a.1.d	– Manual approach, engine(s) inoperative.	X	X	X	X	
7.a.2	Autopilot coupled approach (to a landing or to a go-around)		X	X	X	
7.a.3	GBAS, all engine(s) operating and with one or more engine(s) inoperative.		X	X	X	
7.b	Non-precision approach. (As applicable)					
7.b.1	Surveillance radar approach, all engine(s) operating and with one or more engine(s) inoperative.	X	X	X	X	
7.b.2	NDB approach, all engine(s) operating and with one or more engine(s) inoperative.	X	X	X	X	
7.b.3	VOR, VOR/DME, TACAN approach, all engines(s) operating and with one or more engine(s) inoperative.	X	X	X	X	
7.b.4	RNAV / RNP / GNSS (RNP at nominal and minimum authorized temperatures) approach, all engine(s) operating and with one or more engine(s) inoperative.	X	X	X	X	Must include a capability to activate integrity loss indications
7.b.5	ILS (LOC), all engine(s) operating and with one or more engine(s) inoperative.	X	X	X	X	

Table E3A – Aeroplane Non-Type-Specific FTD - Subjective Requirements

Entry #	Functions and Subjective Tests	Level		Qualification		Comment
		1	2	MCC	IPC	
7.b.6	ILS offset localizer approach, all engine(s) operating and with one or more engine(s) inoperative.	X	X	X	X	
7.c	Approach procedures with vertical guidance (APV), e.g. SBAS, flight path vector.					
7.c.1	APV/baro-VNAV approach, all engine(s) operating and with one or more engine(s) inoperative.	X	X	X	X	Must include a capability to activate of integrity loss indications
7.c.2	Area navigation (RNAV) approach procedures based on SBAS, all engine(s) operating and with one or more engine(s) inoperative.	X	X	X	X	
8. Visual approaches (segment) and landings. (As applicable)						
8.a	Manoeuvring, normal approach and landing all engines operating, with and without visual and navigational approach aid guidance.		X	X	X	
8.b	Approach and landing with one or more engine(s) inoperative.		X	X	X	
8.c	Operation of landing gear, flap/slats and speedbrakes (normal and abnormal).		X	X	X	
8.d	Approach and landing with crosswind (maximum demonstrated crosswind component).		X	X	X	
8.e	Approach and landing with flight control system failures (for reconfiguration modes, manual reversion and associated handling with the most significant degradation which is probable).		X	X	X	
8.f	Approach and landing with standby(minimum) electrical/hydraulic power.			X		If abnormal operation is designated and approved for use.
8.g	Approach and landing from circling conditions (circling approach). Note: requires as a minimum a representative airport scene that can provide a heading difference of 90°, or more, and 180°, or less, between approach and landing runways. Any associated hazard lights or any other visual aids for use as part of the published circling procedure should be included in the correct position(s) and be of the appropriate colour(s), directionality and behaviour. A generic airport model to be consistent with published data used for aeroplane operations may be used and should contain both the approach and landing runways and have the capability to light both at the same time. Any associated hazard lights or any other visual aids for use as part of the published circling procedure need to be included in the correct position(s) and be of the appropriate colour(s) and behaviour.		X	X		Procedural training only. Visual FOV must be 150x30 degrees minimum.
8.h	Approach and landing from visual traffic pattern.		X	X	X	
8.i	Approach and landing from non-precision approach.		X	X	X	
8.j	Approach and landing from precision approach.		X	X	X	

Table E3A – Aeroplane Non-Type-Specific FTD - Subjective Requirements

Entry #	Functions and Subjective Tests	Level		Qualification		Comment
		1	2	MCC	IPC	
8.k	Other. Note: An FTD with a visual system permitting a special approach procedure in accordance with the CARs, may be approved for that particular approach procedure.		X	X	X	
9. Missed Approach.						
9.a	All engines, manual and autopilot.	X	X	X	X	
9.b	Engine(s) inoperative, manual and autopilot.	X	X	X	X	
9.c	Rejected landing.	X	X	X	X	
10. Surface operations (landing, after-landing and post-flight).						
10.a	Landing roll and taxi. (As applicable)					
10.a.1	HUD/EFVS.			X	X	
10.a.2	Spoiler operation.		X	X	X	
10.a.3	Reverse thrust operation.		X	X	X	
10.a.4	Directional control and ground handling, both with and without reverse thrust.	X	X	X	X	
10.a.6	Brake and anti-skid operation with dry conditions.	X	X	X	X	If visual is installed.
10.b	Engine shutdown and parking.					
10.b.1	Engine and systems operation.	X	X	X	X	
10.b.2	Parking brake operation.		X	X	X	
10.b.3	Other. As applicable to intended additional approved training	X	X	X	X	
11. Any flight phase						
11.a	Aeroplane and engine systems operation (Subject to evaluation if installed or required for approved use).					
11.a.1	Air conditioning and pressurisation (Environmental Control System).			X		
11.a.2	De-icing/anti-icing.		X	X	X	
11.a.3	Auxiliary engine/auxiliary power unit (APU).			X		
11.a.4	Communications.	X	X	X	X	
11.a.5	Electrical.	X	X	X	X	
11.a.6	Fire and smoke detection and suppression.			X		
11.a.7	Flight controls (primary and secondary).	X	X	X	X	
11.a.8	Fuel and oil.	X	X	X	X	
11.a.9	Hydraulic.	X	X	X		
11.a.10	Pneumatic.	X	X	X		
11.a.11	Landing gear.	X	X	X	X	
11.a.12	Oxygen.			X		
11.a.13	Engine.	X	X	X	X	
11.a.14	Airborne radar.		X	X		
11.a.15	Autopilot and flight director.	X	X	X	X	
11.a.16	Terrain awareness warning systems and collision avoidance systems (e.g. EGPWS, GPWS, TCAS).	X	X	X		

Table E3A – Aeroplane Non-Type-Specific FTD - Subjective Requirements

Entry #	Functions and Subjective Tests	Level		Qualification		Comment
		1	2	MCC	IPC	
11.a.17	Flight control computers including stability and control augmentation.		X	X		
11.a.18	Flight displays systems.	X	X	X	X	
11.a.19	Flight management systems.	X	X	X	X	
11.a.20	Head-up displays (including EFVS, if appropriate).		X	X		
11.a.21	Navigation systems.	X	X	X	X	
11.a.22	Stall warning/avoidance.	X	X	X	X	
11.a.28	Other.					
11.b	Airborne procedures.					
11.b.1	Holding.	X	X	X	X	
11.b.2	Air hazard avoidance (traffic, weather, including visual correlation).		X	X		If installed
12. Visual System. This section applies to real-world airports models designated by the sponsor for approved use, which will be used to complete the functions and subjective tests described in this Appendix. The airports should service the aeroplane type being simulated, be used regularly in the training programme(s) and, as applicable. However, if the requirement for device level allows, the sponsor may present models for use during the initial qualification which do not fully replicate any particular airport (fictitious airport). During recurrent evaluations the evaluator may select any visual scene used in the operator's training programme(s) for completion of the functions and subjective tests, provided these scenes were modelled with the features required.						
12.a	Functional test content requirements. The following are the minimum airport model content requirements to satisfy visual capability tests, and provide suitable visual cues to allow completion of all functions and subjective tests described in this appendix. FSTD operators are encouraged to use the model content described below for the functions and subjective tests.					
12.a.1	Airport scenes.} A minimum of one (1) generic airport model to be consistent with published data used for aeroplane operations. This model should be acceptable to the NSEP and selectable from the IOS.	X*	X	X		Sponsor must designate real –world scenes for qualification prior to the initial evaluation.
	A minimum of three (3) real-world airport models to be consistent with published data used for aeroplane operations and capable of demonstrating all the visual system features below. Each model should be in a different visual scene to permit assessment of FSTD automatic visual scene changes. Each model should be selectable from the IOS.				X	*If a visual is installed.
12.a.2	Visual scene fidelity. A generic or fictitious airport visual scene with sufficient fidelity for the aircrew to successfully accomplish take-offs, approaches, and landings.	X				If visual system is installed
	The “non-designated” visual scenes must have sufficient fidelity for the aircrew to visually identify the airport; determine the position of the simulated aeroplane; successfully accomplish take-offs, approaches, and landings; and manoeuvre around the airport on the ground as necessary.		X	X		For non-designated scenes

Table E3A – Aeroplane Non-Type-Specific FTD - Subjective Requirements

Entry #	Functions and Subjective Tests	Level		Qualification		Comment
		1	2	MCC	IPC	
	The “designated” visual scenes should correctly represent the parts of the airport and its surroundings used in training or checking.		X	X	X	For designated scenes
12.a.3	Runways and taxiways. Generic runways and taxiways.	X				If visual system is installed
	Representative runways and taxiways.		X	X		For non-designated scenes
	The airport runways and taxiways.		X	X	X	For designated scenes
12.a.4	Runway threshold elevations and locations should be modelled to provide correlation with aeroplane systems (e.g. HUD, GPS, compass, altimeter).	X	X	X	X	
12.a.5	Runway surface and markings for each “in-use” runway should include the following, if appropriate: Threshold markings. Runway numbers. Touchdown zone markings. Edge markings. Centre line markings. Windsock that gives appropriate wind cues.	X	X	X	X	The feature, if required, should be representative for “designated scenes” and generic for “non-designated” scenes.
	Fixed distance markings.		X	X	X	
	Signs at intersecting runways and taxiways.			X	X	
12.a.6	Runway lighting of appropriate colours, directionality, behaviour and spacing for the “in-use” runway including the following: Threshold lights. Edge lights. End lights. Centre line lights. Touchdown zone lights. Lead-off lights. Appropriate visual landing aid(s) for that runway. Appropriate approach lighting system for that runway.	X	X	X	X	The feature, if required, should be representative for “designated scenes” and generic for “non-designated” scenes.
	– Representative airport buildings, structures and lighting.		X		X	
12.a.7	Taxiway surface and markings (associated with each “in-use” runway): Centre line markings. Runway holding position markings.	X	X	X	X	The feature, if required, should be representative for “designated scenes” and generic for “non-designated” scenes.
	Edge markings ILS critical area markings.		X	X	X	
	A including all taxiway markings, lighting, and signage to taxi, as a minimum, from a designated parking position to a designated runway and return, after landing on the designated runway, to a designated parking position. The designated runway and taxi routing should be consistent with that airport for operations in low visibilities.		X	X		Low visibility taxi route applies only if low visibility procedural training is intended.

Table E3A – Aeroplane Non-Type-Specific FTD - Subjective Requirements

Entry #	Functions and Subjective Tests	Level		Qualification		Comment
		1	2	MCC	IPC	
12.a.8	Taxiway lighting of appropriate colours, directionality, behaviour and spacing (For each “in-use” runway): Edge lights. Centre line lights. Runway holding position and ILS critical area lights.	X	X	X	X	The feature, if required, should be representative for “designated scenes” and generic for “non-designated” scenes.
12.a.9	Required visual model correlation with other aspects of the airport environment simulation. The airport model should be properly aligned with the navigational aids that are associated with operations at the runway “in-use”.	X	X	X	X	
12.a.10	Airport buildings, structures and lighting. Representative airport buildings, structures and lighting.		X		X	
	Generic airport buildings, structures and lighting.	X		X		
12.a.11	Terrain and obstacles Representative depiction of terrain and obstacles within 46 km (25 NM) of the reference airport.		X	X		
12.a.12	Significant, identifiable natural and cultural features. Representative depiction of significant and identifiable natural and cultural features within 46 km (25 NM) of the reference airport. (See Note) Representative moving airborne traffic (including the capability to present air hazards – e.g. airborne traffic on a possible collision course). (if installed)		X	X		Note: This refers to natural and cultural features typically used for pilot orientation in flight.
12.b	Visual scene management. Airport runway, approach and taxiway lighting and cultural lighting intensity for any approach should be set at an intensity representative of that used in training for the visibility set; all visual scene light points should fade into view appropriately.	X	X	X	X	
12.c	Visual feature recognition. Note: The following are the minimum distances at which runway features should be visible. Distances are measured from runway threshold to an aeroplane aligned with the runway on an extended 3-degree glide slope in suitable simulated meteorological conditions. For circling approaches, all tests below apply both to the runway used for the initial approach and to the runway of intended landing.					
	Runway definition, strobe lights, approach lights, and runway edge white lights from 8 km (5 sm) of the runway threshold. Visual approach aids lights from 4.8 km (3 sm) of the runway threshold.					
	Runway centre line lights and taxiway definition from 4.8 km (3 sm). Threshold lights and touchdown zone lights from 3.2 km (2 sm). For circling approaches, the runway of intended landing and associated lighting should fade into view in a non-distracting manner.	X	X	X	X	

Table E3A – Aeroplane Non-Type-Specific FTD - Subjective Requirements

[illegible]

Table E3A – Aeroplane Non-Type-Specific FTD - Subjective Requirements

Entry #	Functions and Subjective Tests	Level		Qualification		Comment
		1	2	MCC	IPC	
16.a	Repositions: Ramp/gate. Take-off. Approach.	X	X	X	X	repositions should be in-trim at the appropriate speed and configuration for the point
16.b	Resets: System. Temperature. Fluids and agents.	X	X	X	X	
16.c	Environment:					
16.c.1	Weather pre-sets: Unlimited, CAVOK, VFR, non-precision, precision (CAT I), (if appropriate).		X		X	
	Unlimited, CAVOK, VFR.	X		X		
16.c.2	Visual effects: Time of day (day, dusk, night); clouds (bases, tops) in feet; visibility in statute miles and RVR in feet. Note: IOS inputs should be tuned as follows: Visibility - to see the runway threshold environment (no lights), RVR - to attain visual contact with runway threshold lights.	X	X	X	X	
16.c.3	Wind: Surface. Turbulence. Surface temperature Atmospheric pressure (QNH).	X	X	X	X	
16.d	Airport: Active runway selection. Airport lighting.	X	X	X	X	
16.e	Aeroplane configuration (fuel, weight, cg, etc.).	X	X	X	X	
16.f	FMS - reloading of programmed data unless precluded by installed equipment.	X	X	X	X	
16.g	Plotting and recording (take-off and approach).		X	X	X	
16.h	Malfunctions (inserting and removing).	X	X	X	X	

Attachments to Chapter 6

Attachment 6-1 – Sample FTD Certificate for Regional Authorities

 Transport Canada		 Transports Canada	
CERTIFICATE FLIGHT TRAINING DEVICE (FTD)		TC ID 4911	CERTIFICAT DISPOSITIF D'ENTRAÎNEMENT DE VOL (DEV)
Sponsor		Commanditaire	
Roadrunner Flight Academy			
Location		Endroit	
Prairie Ville, SK, Canada			
Manufacturer / Model		Constructeur / Modèle	
Simulo Volitus IX			
Aircraft Class/Set		Classe/Ensemble d'aéronef	
Twin-Engine Turboprop			
Serial Number		Numéro de série	
YLE-Ky0T3			
<i>This certificate is issued pursuant to Section 606.03 of the Canadian Aviation Regulations.</i> <i>Transport Canada – Prairie and Northern Region has evaluated the above Flight Simulation Training Device and found it to comply with:</i> Level 2-IPC/MCC <i>qualification standards and criteria, based on:</i> <i>Dated at Winnipeg, Canada</i>		<i>Ce certificat est délivré en vertu de la section 606.03 du Règlement de l'aviation canadien.</i> <i>Transports Canada – Région des Prairies et du Nord a évalué le dispositif d'entraînement de vol et a jugé qu'il était conforme à :</i> Niveau 2-IPC/CE <i>à la norme et aux critères de qualification, selon:</i> <i>Daté à Winnipeg Canada</i>	
		For the Minister of Transport	
		Pour le Ministre des Transports	
		Valid Until : Valide jusqu'au :	
		2019-01-31	
This qualification is provisional and may be suspended if the Operator does not satisfactorily address the findings documented in the FSTD Evaluation Report upon which this Certificate was based.		Cette qualification est provisoire et peut être suspendue si l'exploitant ne donne pas suite de façon satisfaisante aux constatations formulées dans le rapport d'évaluation FSTD sur lequel repose le présent certificat.	
RDIMS/SGDDI#			

Attachment 6.2 - Non-Type-Specific FTD Qualification Profile

Attachment 6.2 - Non-Type-Specific FTD Qualification Profile			
FTD Information		Sponsor Information	
Sponsor:	Roadrunner Flight Academy		
TCID Number:	4911		
FTD Location:	Prairie Ville, SK		
FTD Model:	Simulo Volitus IX		
Aircraft Class/Set:	Twin-Engine Turboprop		
Qualification Level:	Level 2-IPC/MCC		
Qualification Standard:	TP9685 Rev3		
FTD Manufacturer:	ACME FTD		
Date of Manufacture:	2018		
Serial Number:	YLE-Ky0T3		
Convertible to:	TCID 4912		
Number of Seats	3		
Simulator Systems		Functional Aircraft Systems	
Visual	Image Gen:	ACME See-right	
	FOV:	150 x 40	
	Projectors:	3 x Spotlight VL – Direct projection	
Computer	ACME PC		
Motion	N/A		
Controls	ACME feels-rite		
Reference Data Record		Approach Qualification	
Precision		Non-Precision	
		RNP APCH	

Attachment 6.2 - Non-Type-Specific FTD Qualification Profile											
Aero-model		Document: ACME Aero D7772									
Engine model		Document: ACME Eng D7772									
Additional Qualifications											
X	IPC		IFR (5 hrs)	X	IFR (10 hrs)	X	MCC				
Other		VFR cross country									
Designated Visual Scenes											
CYBR			CYWG			CYPG					
Designated Abnormals (for MCC)											
Minimum-Electrics Landing						Fuel Leak					
Pressurization Loss											
Manual Gear lowering.											
QTG File Location:				RDIMS #: 139662 (or "held on site")							
RDR File Location:				RDIMS #: 139662							
X	ILS CAT I			X	VOR / DME			X	LNAV		
	GLS			X	NDB			X	LNAV/VNAV		
				X	GNSS			X	LPV		
Qualification Restrictions:											
Manoeuvres, procedures, tasks, and functions not approved due to technical or operational limitations.											
No RNP APCH authorized. Integrity warning inoperative.											